Advanced Modeling & Simulation (AMS) Seminar Series
NASA Ames Research Center, September 29th, 2022

CFD Methods in Aircraft Icing Simulations

Isik Ozcer

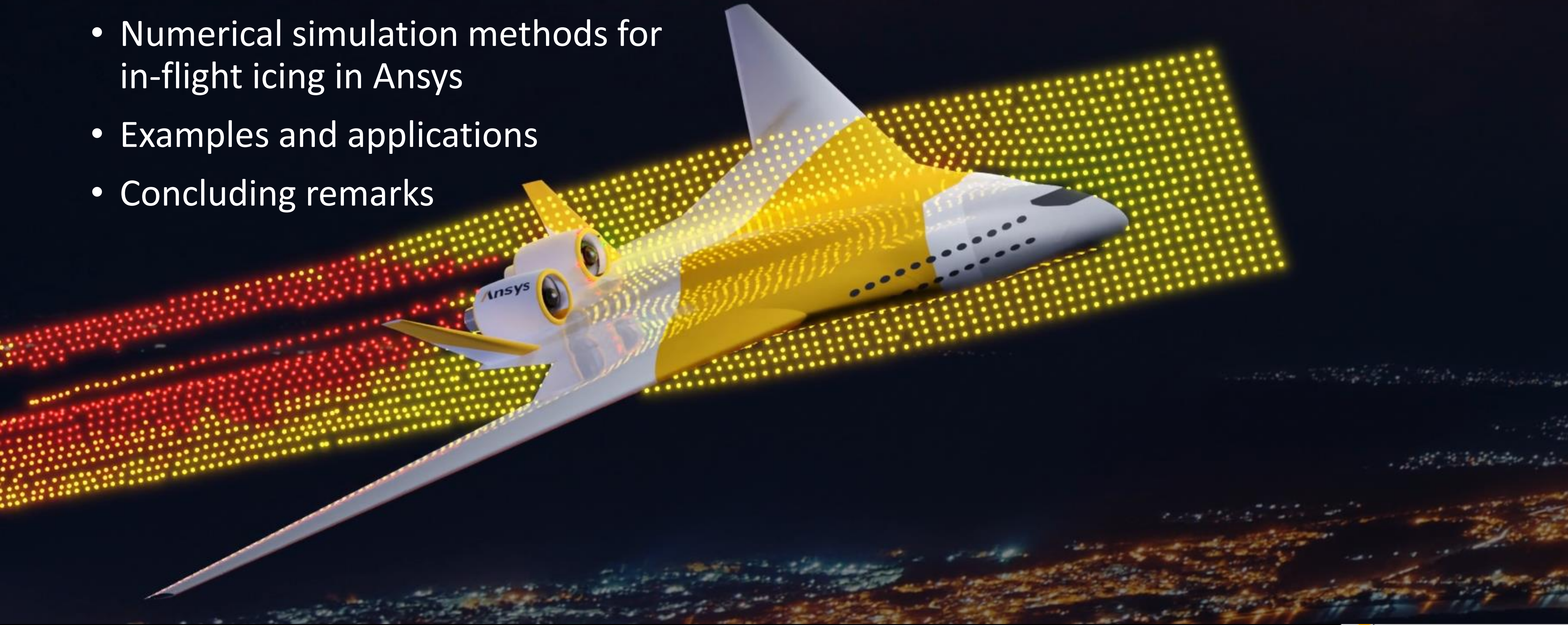
Principal R&D Engineer

Ansys Inc.

Ansys

Agenda

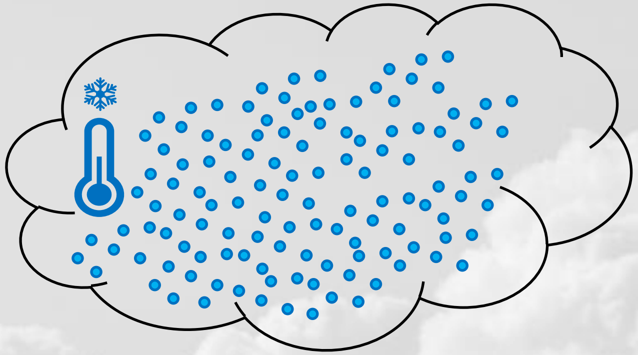
- In-flight icing fundamentals
- Numerical simulation methods for in-flight icing in Ansys
- Examples and applications
- Concluding remarks

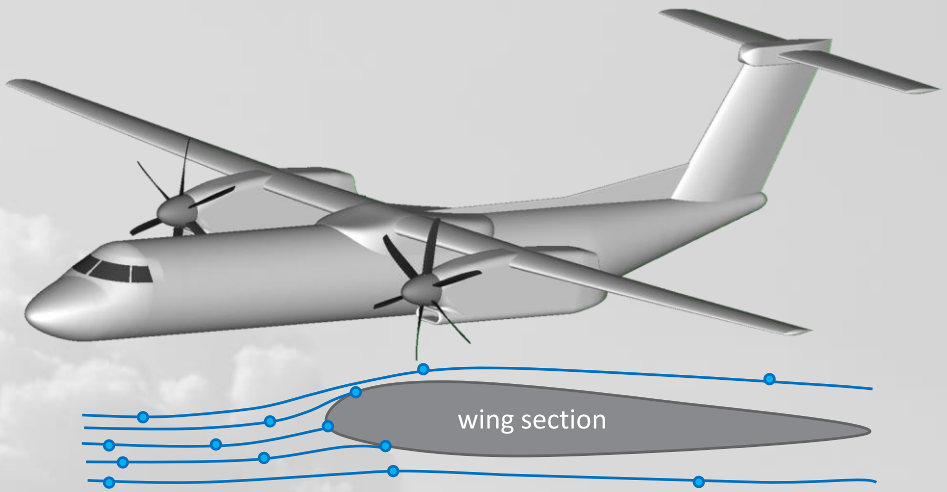


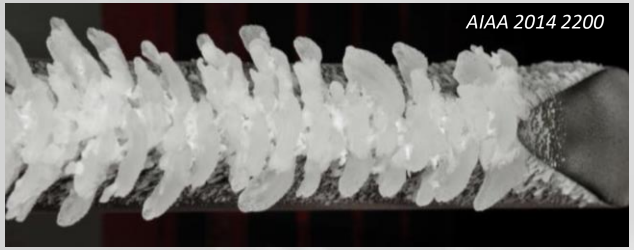
In-flight icing fundamentals

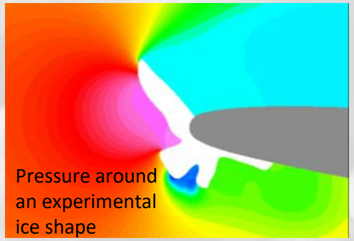
In-flight icing fundamentals

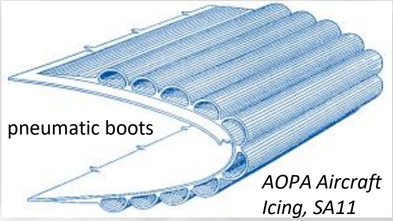
- Icing clouds contain **supercooled** water droplets (**liquid**) below freezing point
- Droplet are tiny, diameters in the order of 10-100 µm (human hair ~ 70 µm)
- When aircraft enter these clouds, their frontal surfaces come into contact with these droplets
- Upon impact, the droplets partly or fully freeze, adhering to surfaces as ice
- As more droplets hit, ice builds-up, large ice shapes can form on wings and tails
- Ice is very strong, does not usually break away with aerodynamic forces
- Iced components begin losing performance – wings, tails, engines, propellers, air speed indicators, etc...
- Aircraft certified to fly into known icing conditions are equipped with Ice Protection Systems

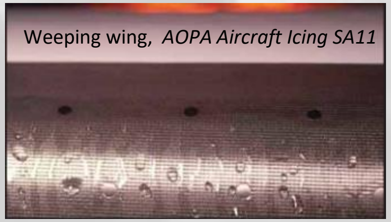


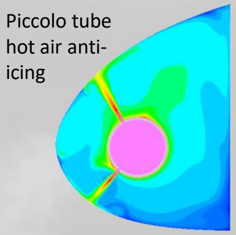










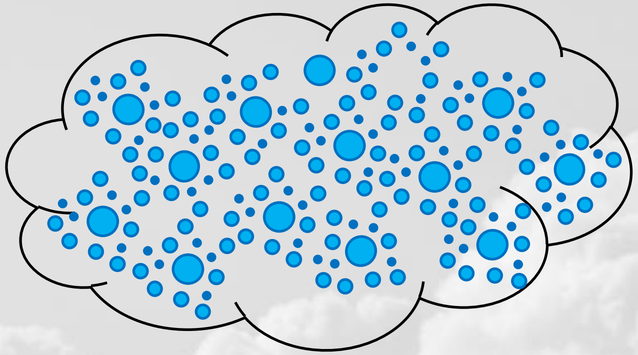


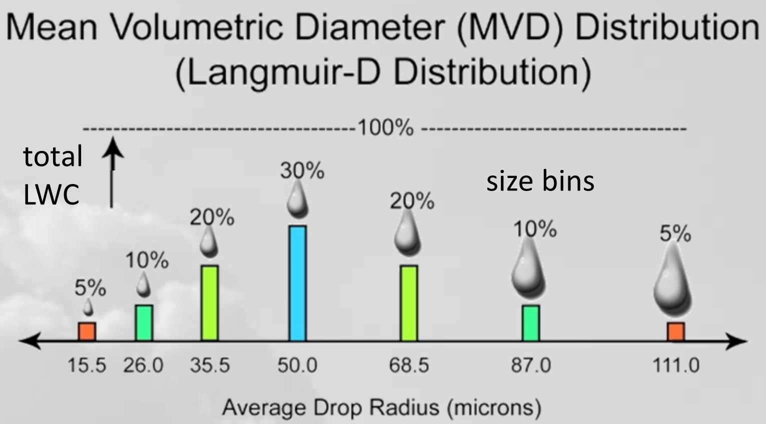
In-flight icing fundamentals

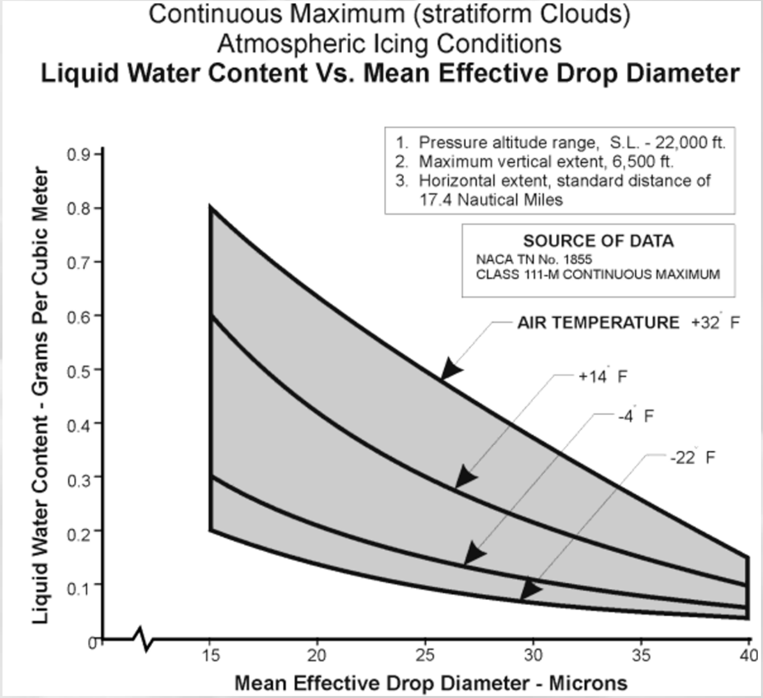
Ice accretion during an experiment at the Icing Research Tunnel - Glenn Research Center, NASA
Video curtesy of Aircarft Owners and Pilots Association

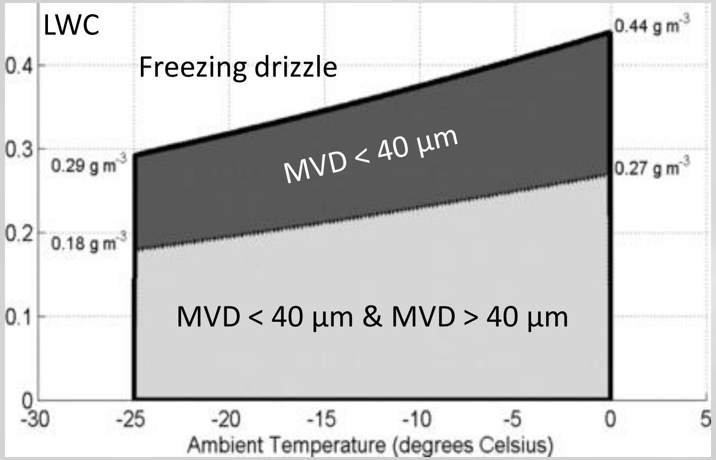
In-flight icing fundamentals

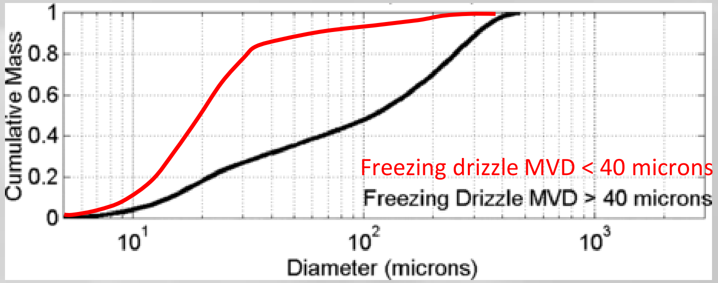
- Icing clouds come in different shapes and forms
- Liquid Water Content – **LWC** (g/m^3): amount of water mass in the cloud
- Particle Size Distribution (**PSD**): Droplets in a cloud have various sizes, which are grouped in “bins”
- Mean volumetric diameter (**MVD**): An average size to represent the icing cloud: 50% of LWC is stored in droplets at or below this size
- Two clouds with same MVD may be very different - **PSD matters**!
- **Appendix C -** FAR 25: Icing design envelope since 1964, MVD 15-50 µm
- Supercooled Large Droplets (**SLD**): Diameter range: 50 – 5000 µm
  - MVD insufficient to describe clouds with SLD
  - SLDs can be very dangerous
  - Included as **App O** in FAR 25 as of Jan 2015



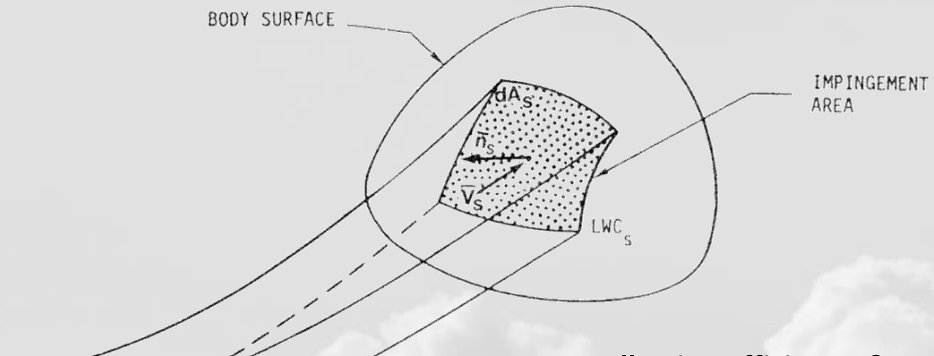








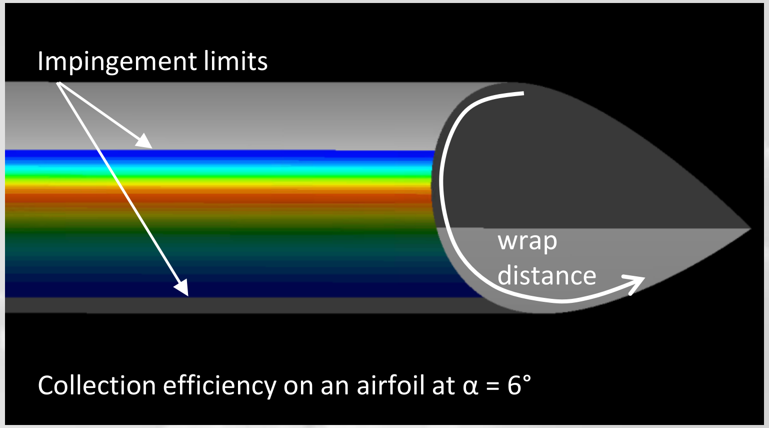
In-flight icing fundamentals

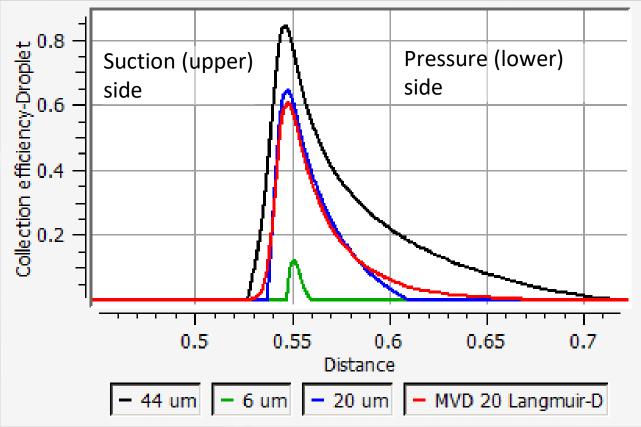


NASA-TM-2002-211700

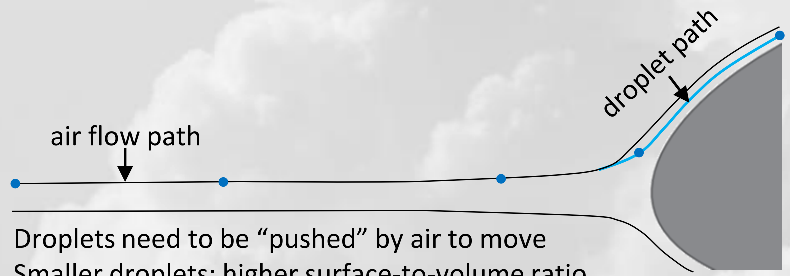
Collection efficiency, β:

$$\beta = \frac{dA_\infty}{dA_s} = \frac{LWC_s\left(-\bar{n}_s \cdot \bar{V}_s\right)}{LWC_\infty V_\infty}$$



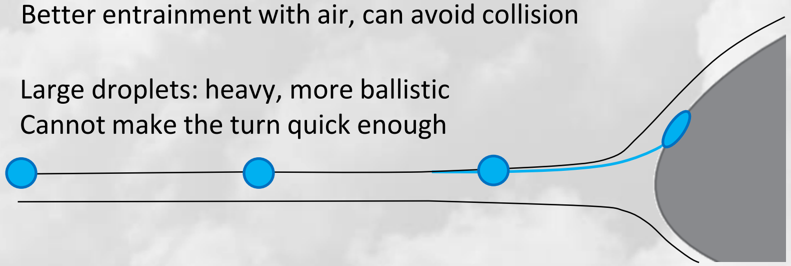


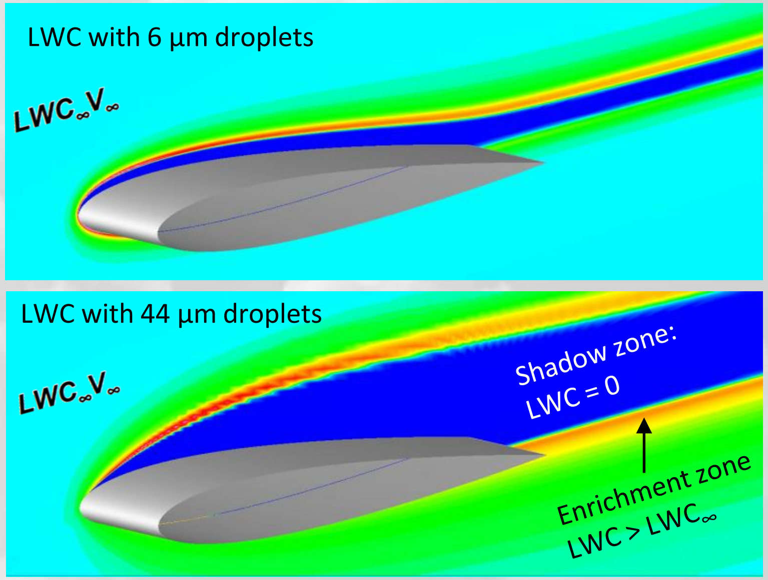
Collection efficiency with different droplet sizes

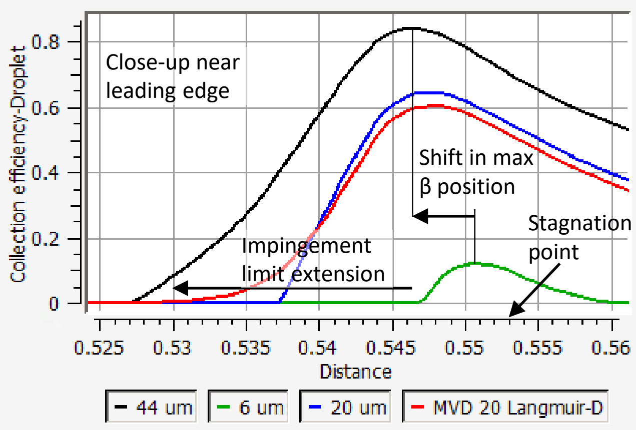


Droplets need to be “pushed” by air to move
Smaller droplets: higher surface-to-volume ratio
Better entrainment with air, can avoid collision

Large droplets: heavy, more ballistic
Cannot make the turn quick enough







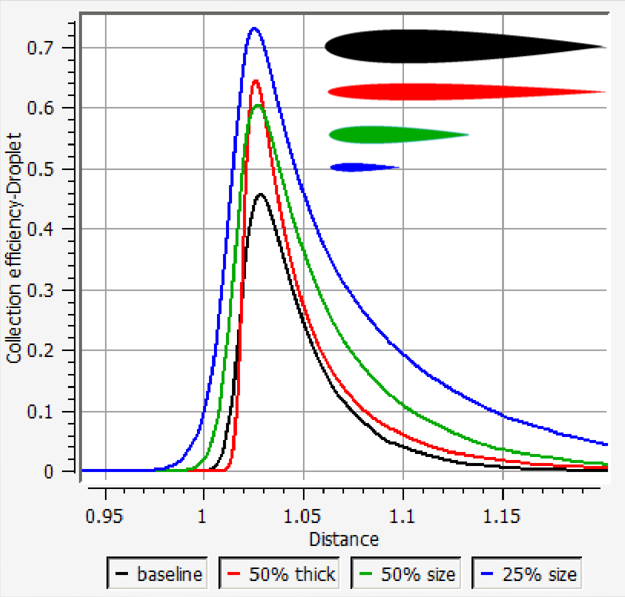
! Stagnation point is different than max β location !

! β > 1 is possible for surfaces in the wake of enrichment !

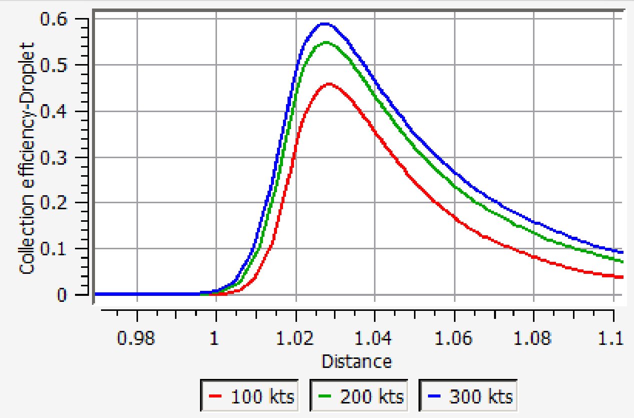
Ansys

In-flight icing fundamentals

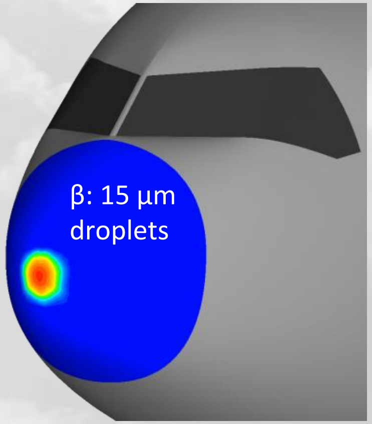
- Primary factors that affect collection efficiency, β
  - Droplet size: most relevant during flight
  - Flight speed
  - Aircraft size
- Secondary factors
  - Air density,
  - Temperature,
  - Turbulence
  - Wakes and enrichment zones, …
- Icing risks increase with reduction in size:
  - Probes and sensors > Propellers and engines > Tails > Wings
  - Drones, UAVs > GA aircraft > Commercial transport
- Particle size distribution should not be ignored
  - Most App-C cases: Max & total β is similar between PSD and single-size droplet simulations
  - But some cases can show significantly larger β with a PSD simulation

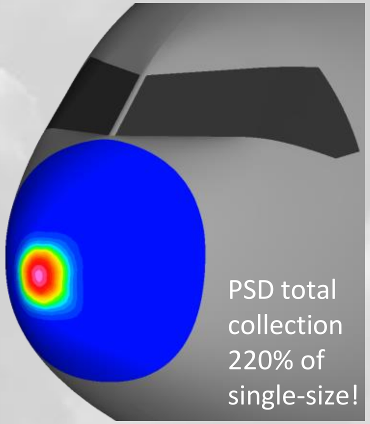


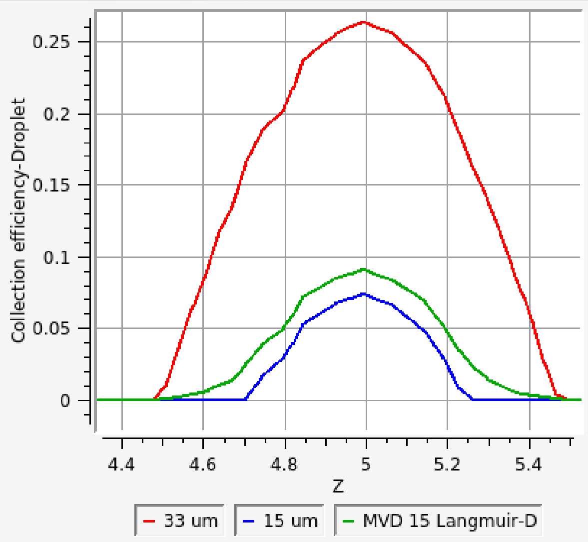
Collection efficiency vs airfoil size, α = 6°
Shown with normalized wrap distance



Collection efficiency vs flight speed, α = 6°







Collection efficiency of a 15 μm cloud on the radome of a full-scale jetliner

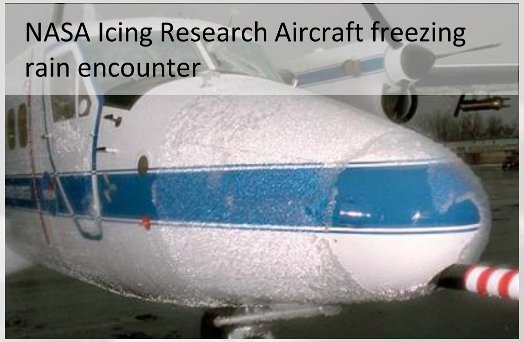
In-flight icing fundamentals

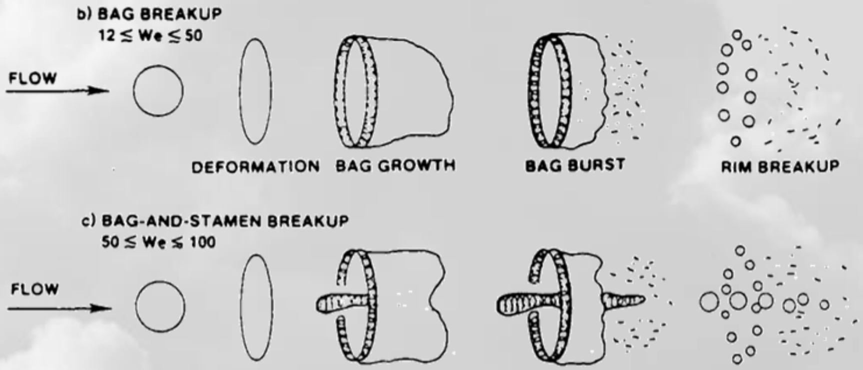
- Supercooled Large Droplets (SLD)
  - Freezing rain/drizzle conditions
  - Droplets much larger than App C conditions, d >> 40 µm
  - Increased collection efficiency and farther reach
  - Very dangerous, especially for small aircraft
  - Always defined by a PSD, MVD not descriptive enough
  - 27 or more droplet size bins to describe an SLD cloud

- Droplet deformation
  - SLDs can deform and break-up with viscous forces
  - Break-up types associated with Weber numbers

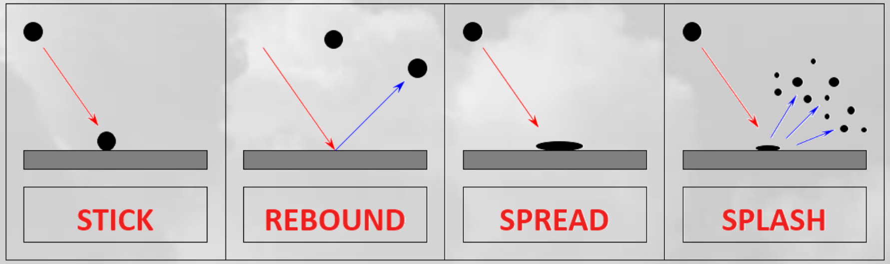
$$We = \frac{\rho_{air}\left|\vec{V}_{air} - \vec{V}_d\right|^2 D}{\sigma_d} \geq 12$$

- Droplet/wall interactions
  - SLDs have complex interactions with aircraft surfaces
  - They don’t readily stick and stay in place
  - They can rebound, spread, and splash (partly stick)
  - Bounced/splashed mass may need to be considered for re-impingement on downstream components



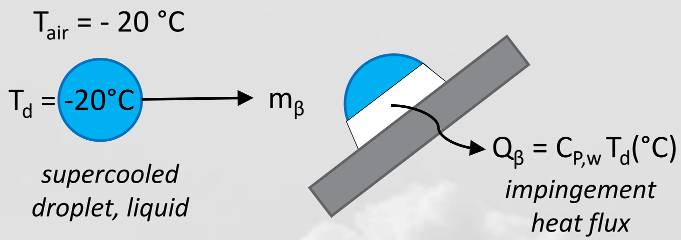


Pilch and Erdman, 1987



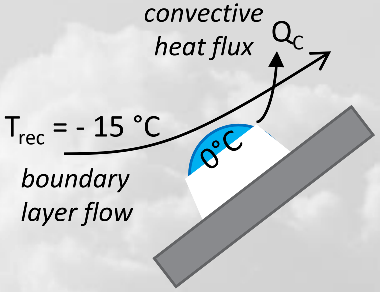
In-flight icing fundamentals

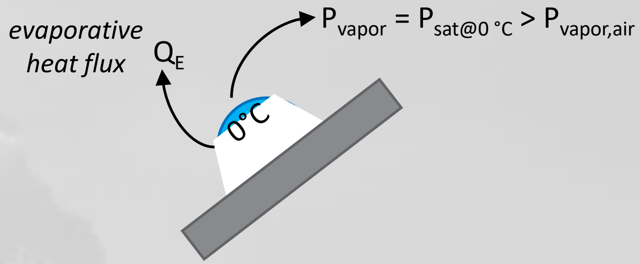
Ice accretion process



$C_{P,w}$ = 4.2 kJ/kg/°C
Q_β = -84 kJ/kg
L_{Fusion} = 334 kj/kg

Q_β, "supercoolness" of droplet only freezes 1/4th and it is not supercool anymore → 0°C



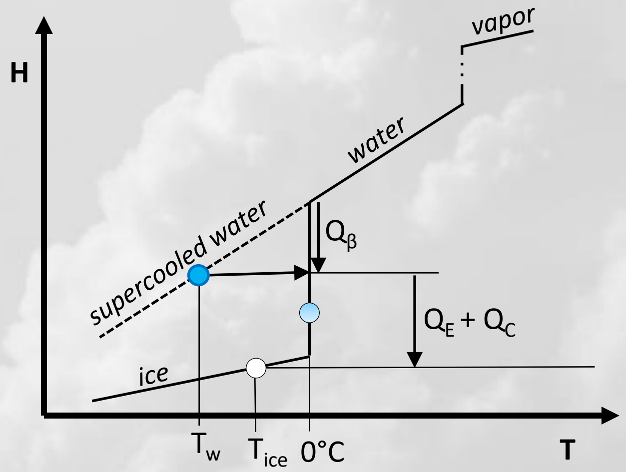


$$-(Q_\beta + Q_C + Q_E) = m_{ice}\,(L_{Fusion} - C_{P,ice}\,T_{ice}(°C))$$

Rime ice $-(Q_\beta + Q_C + Q_E) > m_\beta L_{Fusion}$: all frozen

Glaze ice $-(Q_\beta + Q_C + Q_E) < m_\beta L_{Fusion}$: some m_{water}@0°C remains: Runback!

Runback water moves by:
Air shear & pressure gradient, centrifugal force, gravity, spreading force, capillary forces



Q_C = HTC (T_{rec} - [T_w,T_{ice}])

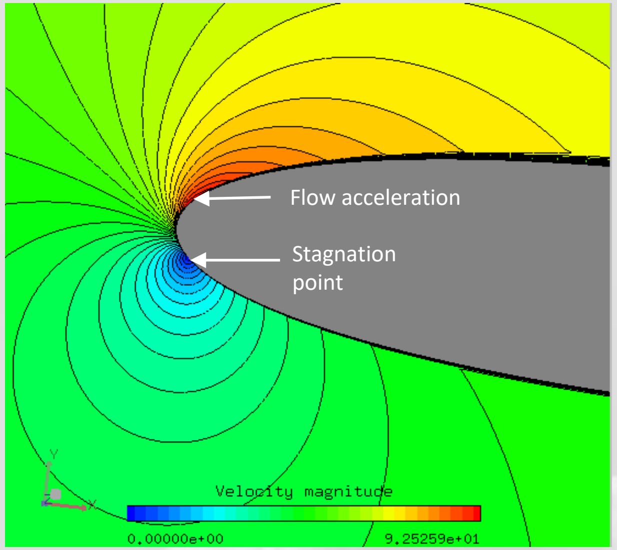
HTC: Heat transfer coefficient

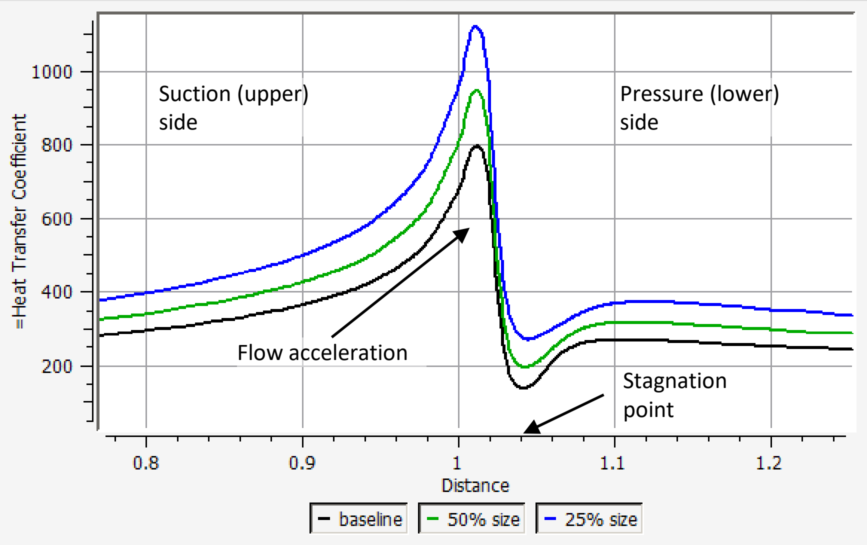
Q_E : HTC & correlation, vapor transport, or both

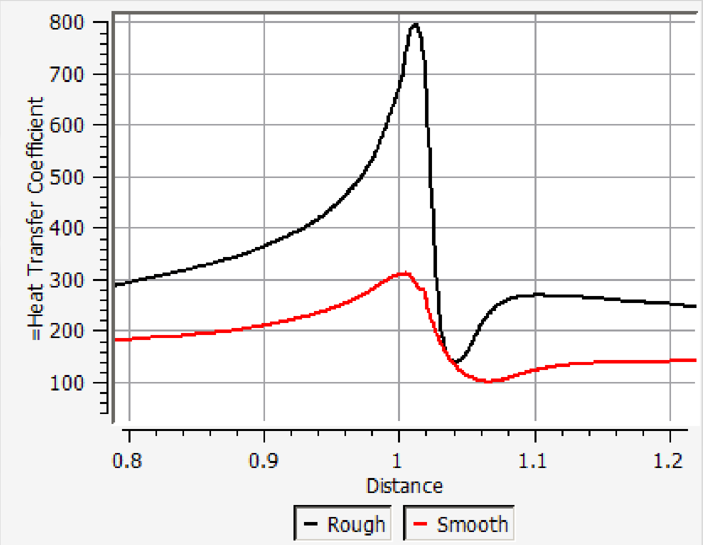
➡ HTC is the key in icing analysis!!!

Ansys

In-flight icing fundamentals





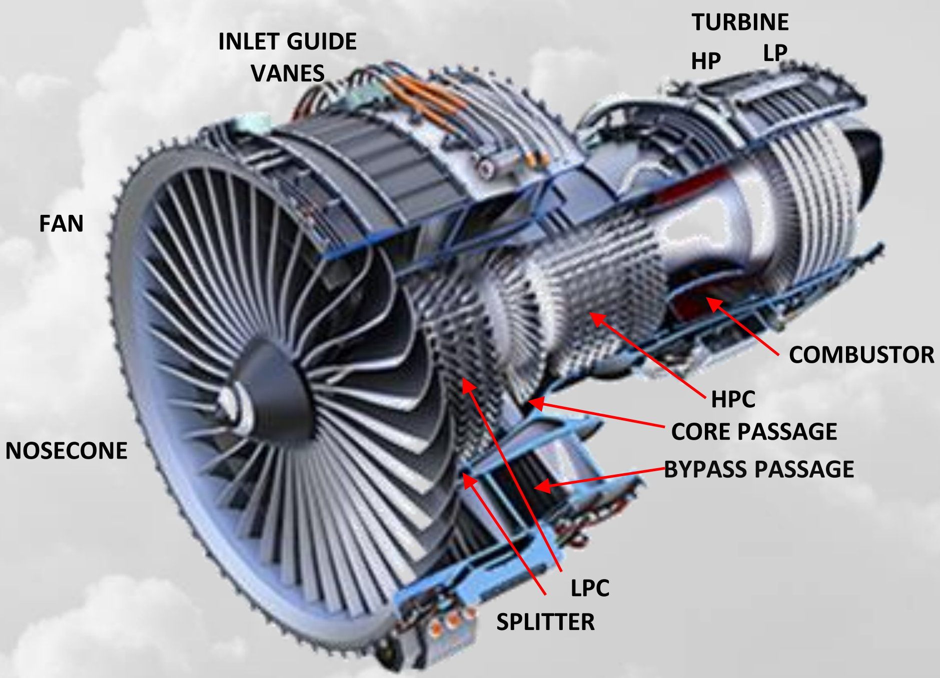


HTC for different size airfoils, α = 6°

Ice is rough. Roughness is not uniform. It changes with time

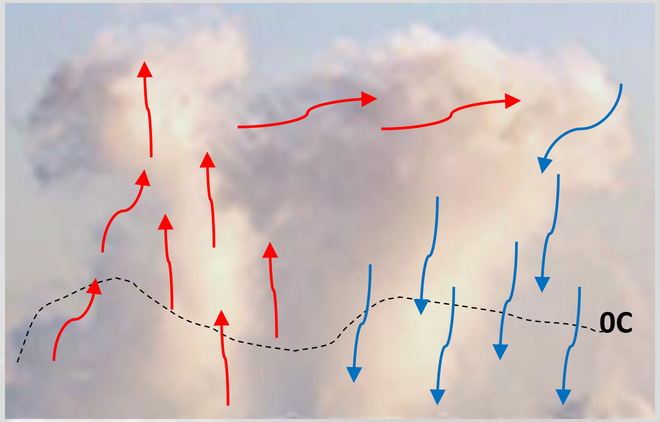
In-flight icing fundamentals

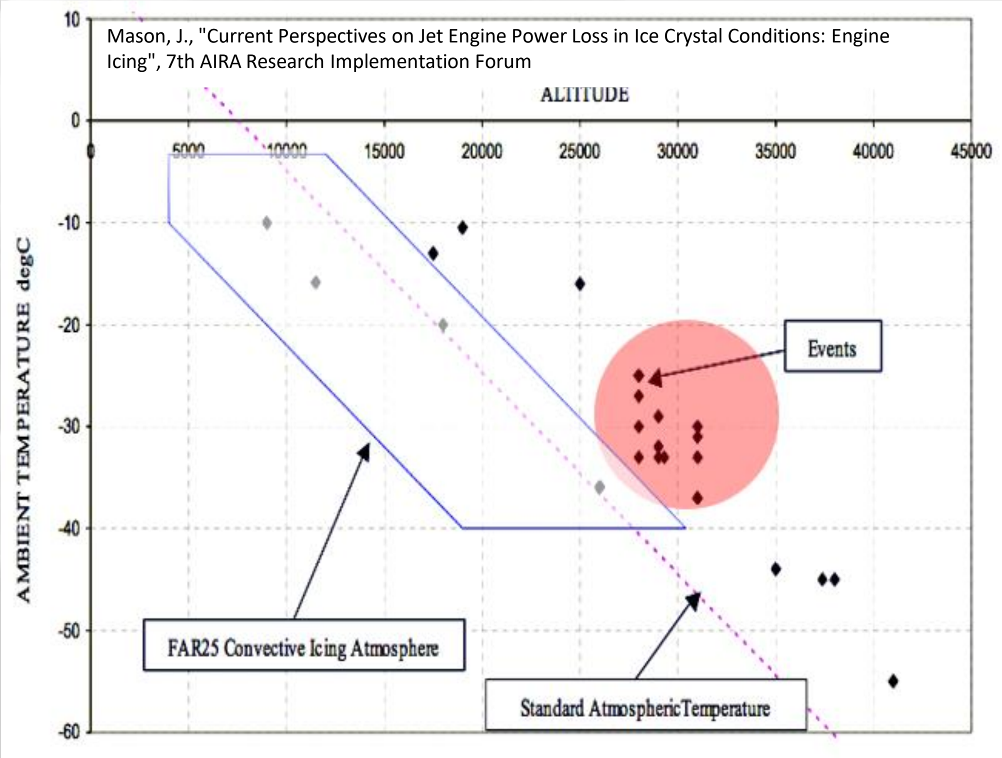
- Droplet icing can happen on all frontal components of an engine:
  - Nacelle lip
  - Sensors
  - Nosecone
  - Fan and first stage compressor
  - Core/bypass splitter lip
  - Inertial particle separators
  - screens ...
- Engine parts are much smaller in size than wings
  - They accrete ice quicker
  - Engine icing: matter of seconds,
    Wing icing: matter of minutes
- Icing and shedding of fan blades can lead to serious vibrations
- Blockage of passages can lead to reduction in engine power
- Rotational effects
  - Aerodynamic heating of blades from root to tip
  - Centrifugal forces on ice and water-film
  - Ice shedding and breakup
  - All such effects should be considered for a complete numerical model



In-flight icing fundamentals

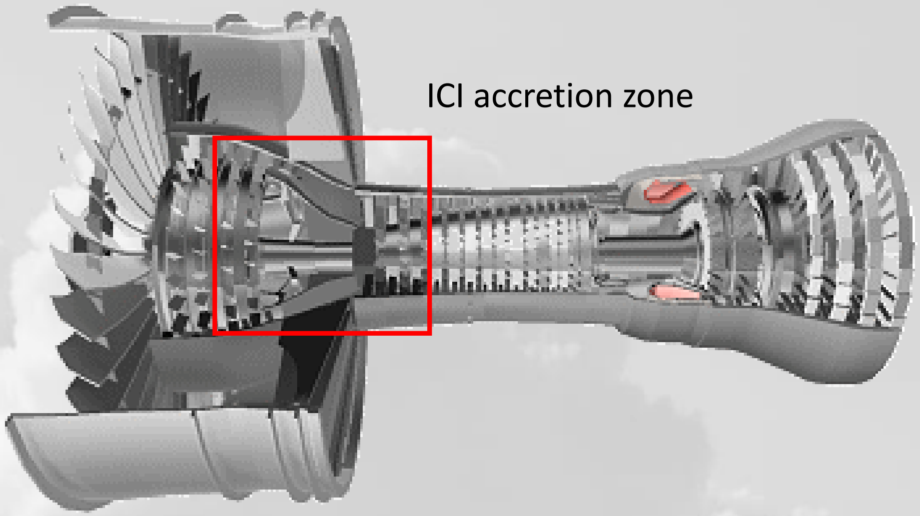
- Ice crystal icing (**ICI**), a hidden threat
  - Clouds with **solid** particles
  - Occurs at high altitude, outside of **App-C** envelope
  - Tropical regions, glaciated clouds with strong convection
  - Low radar visibility of cloud
  - Crystals bounce off cold airframe surfaces, no visible icing on wings/tails
  - Frontal engine surfaces remain ice-free
  - Typical issue: **engine power-loss during cruise** and/or initial descent

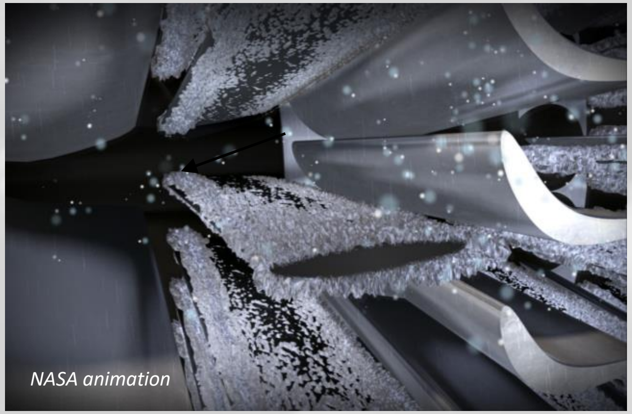


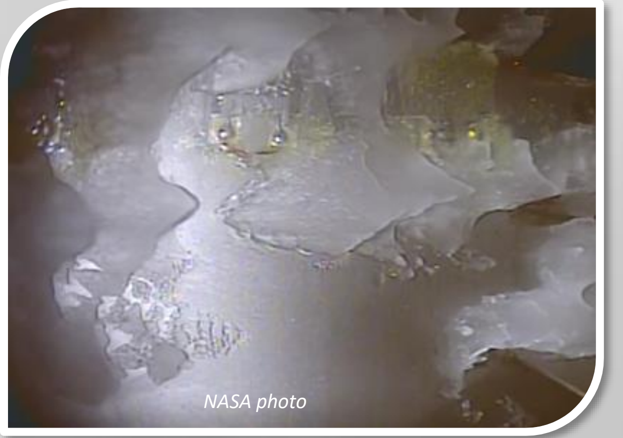


In-flight icing fundamentals

- ICI is a melting problem
  - Crystals melt and stick to deeper engine surfaces
  - Air and metal surface temperatures > 0°C
  - Melting encourages more sticking, rapidly increasing crystal ice growth
  - Ice is **slushy and runny**, unlike glaze or rime ice observed in droplet icing
  - Unstable ice in upstream stages can shed and block narrower downstream passages
- ICI physical aspects for modeling
  - Melting as traversing warm air
  - Sticking and bouncing on surfaces
  - Fragmantation and reinjection
  - Erosion



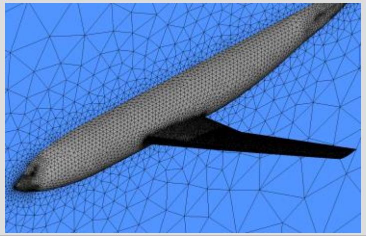




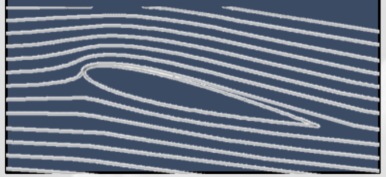
Numerical simulation methods for in-flight icing in Ansys

Ansys in-flight icing simulation methodology overview

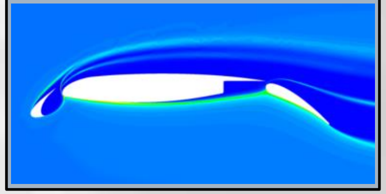
Initial CAD/Mesh, unstructured



Air flow: RANS with roughness: HTC + wall shear



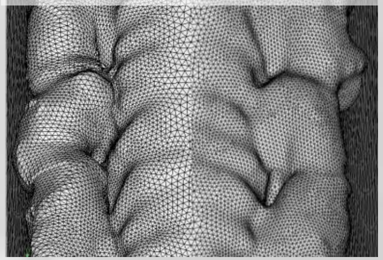
Droplet flow: Eulerian with size distributions



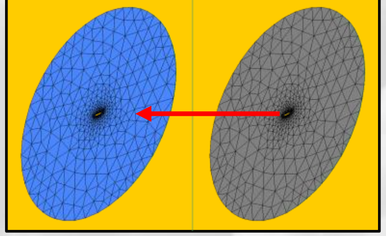
1 shot of icing simulation

Wall film runback and icing: Eulerian, ice roughness and accretion

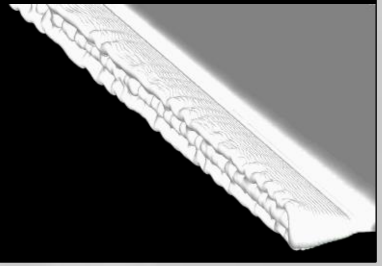
Remeshing iced geometry



Boundary/initial condition update



Aero analysis, 3D printing



Multishot iterations:

Fully automated
<u>Same grids for all solvers</u>
User-free remeshing

Airflow and droplet solvers

- Airflow
  - 3D finite-volume and/or finite element
  - **Fluent, CFX, FENSAP** (FE)
  - RANS or URANS
    - To analytically obtain smooth and rough HTC
  - Turbulence**: S-A, kw-SST**, possibly others
- Droplets and crystals
  - 3D Finite element, **DROP3D**
    - Main tool for icing impingement analysis
    - Same discretization / time integration as FENSAP
    - Eulerian (continuum) formulation
    - Particle thermal energy
    - Vapor transport
    - Two-way thermal coupling with the airflow solver (FENSAP)
  - Fluent discrete phase model (**DPM**)
    - Lagrangian particle tracking
    - Newly being introduced to icing workflow in **Fluent Icing**
    - Intended for secondary effects: reinjection, breakup, fragmentation
    - User defined functions (**UDF**) for SLD and crystal bounce modeling, particle drag relationships
- Both systems support rotating frames for propeller, engine analysis
- Couplings
  - Icing cloud volume fraction is ~ 1e-6, momentum 1-way
  - Vapor, air, particle energy cross coupling
  - Vapor, particle mass coupling

Airflow equations

$$\frac{\partial \rho_a}{\partial t}+\vec{\nabla}\cdot\left(\rho_a\vec{V}_a\right)=0$$

$$\sigma^{ij}=-\delta^{ij}p_a+\tau^{ij}$$

$$\frac{\partial \rho_a\vec{V}_a}{\partial t}+\vec{\nabla}\cdot\left(\rho_a\vec{V}_a\vec{V}_a\right)=\vec{\nabla}\cdot\sigma^{ij}$$

$$\frac{\partial \rho_a E_a}{\partial t}+\vec{\nabla}\cdot\left(\rho_a\vec{V}_a H_a\right)=\vec{\nabla}\cdot\left(\kappa_a\left(\vec{\nabla}T_a\right)+v_i\tau^{ij}\right)$$

$k = k_{eff} = k_{lam} + k_{turb}$

HTC with roughness

Particle flow equations

$$\frac{\partial \alpha}{\partial t}+\vec{\nabla}\cdot\left(\alpha\vec{V}_d\right)=S_\alpha$$

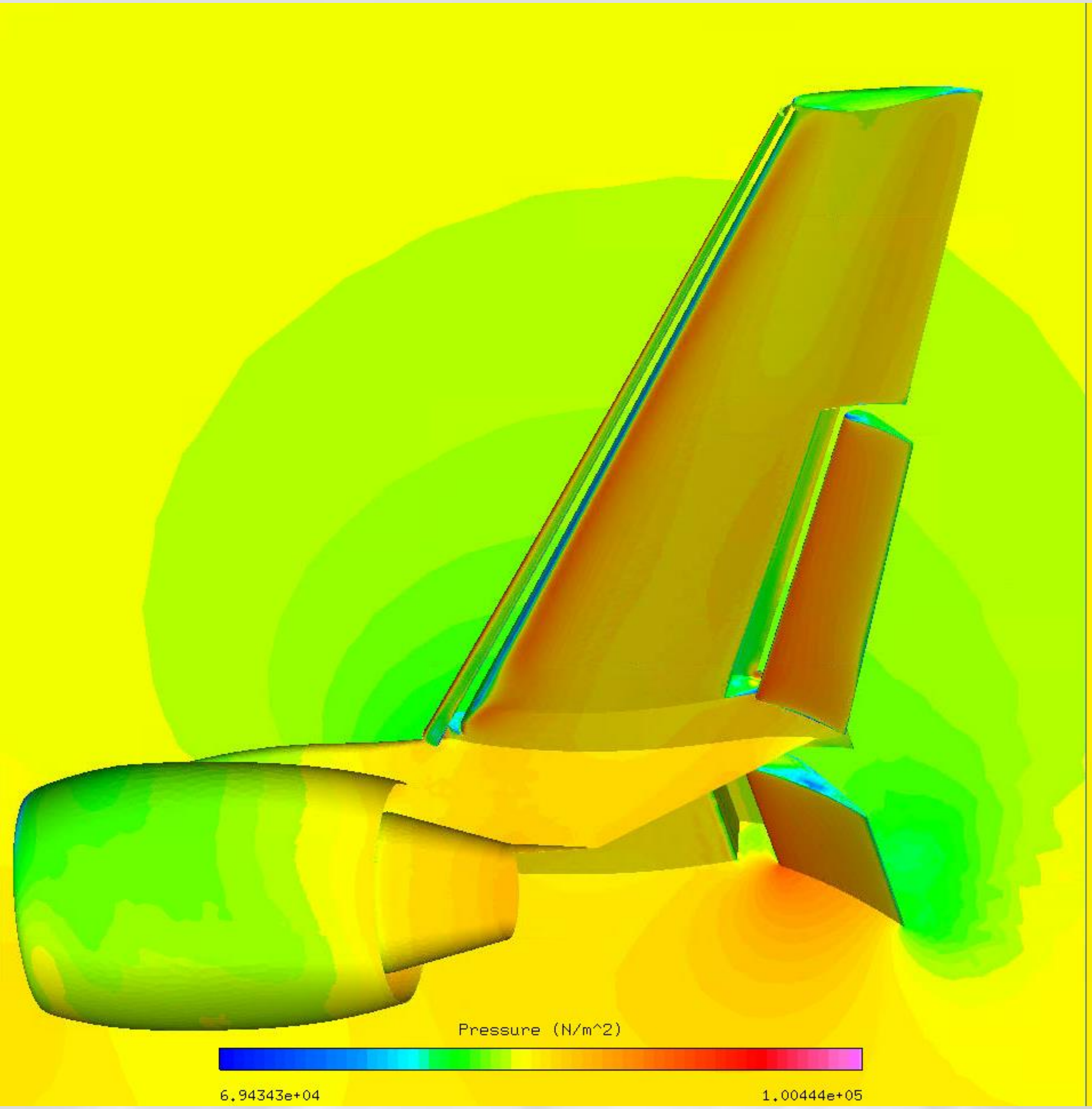
$$\frac{\partial\left(\alpha\vec{V}_d\right)}{\partial t}+\vec{\nabla}\cdot\left[\alpha\vec{V}_d\,\vec{V}_d\right]=\frac{C_D\mathrm{Re}_d}{24K}\alpha\left(\vec{V}_a-\vec{V}_d\right)$$

$$\frac{\partial E}{\partial t}+\frac{\partial\left(v_jE\right)}{\partial x_j}=\alpha\left(\dot{q}_{conv}+\dot{s}_{rad}+\dot{q}_{evap/cond}+\dot{q}_{freeze/melt}\right)$$

$E=\alpha i$

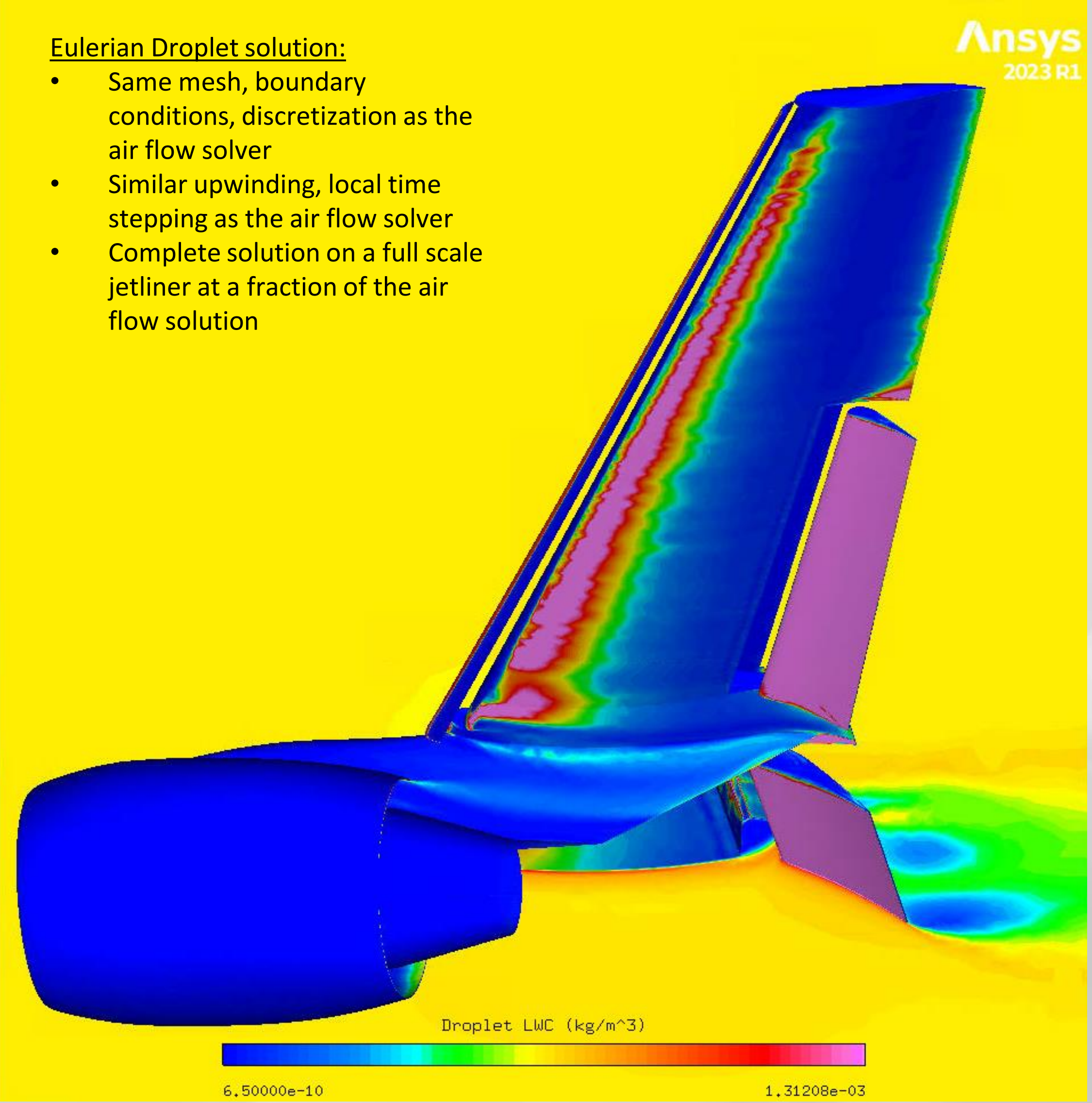
$$\frac{\partial \alpha_{vpr}}{\partial t}+\vec{\nabla}\cdot\left(\alpha_{vpr}\overrightarrow{V_{r,a}}\right)=\vec{\nabla}\cdot\left(D_{eff}\nabla\phi\right)+S$$

Airflow and droplet solvers

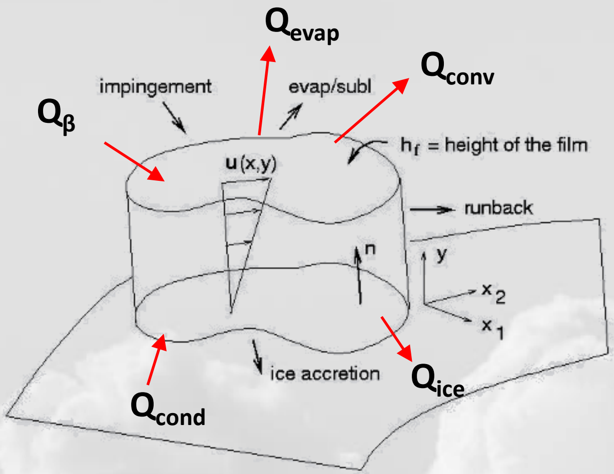


Eulerian Droplet solution:

- Same mesh, boundary conditions, discretization as the air flow solver
- Similar upwinding, local time stepping as the air flow solver
- Complete solution on a full scale jetliner at a fraction of the air flow solution



Ice accretion solver: ICE3D

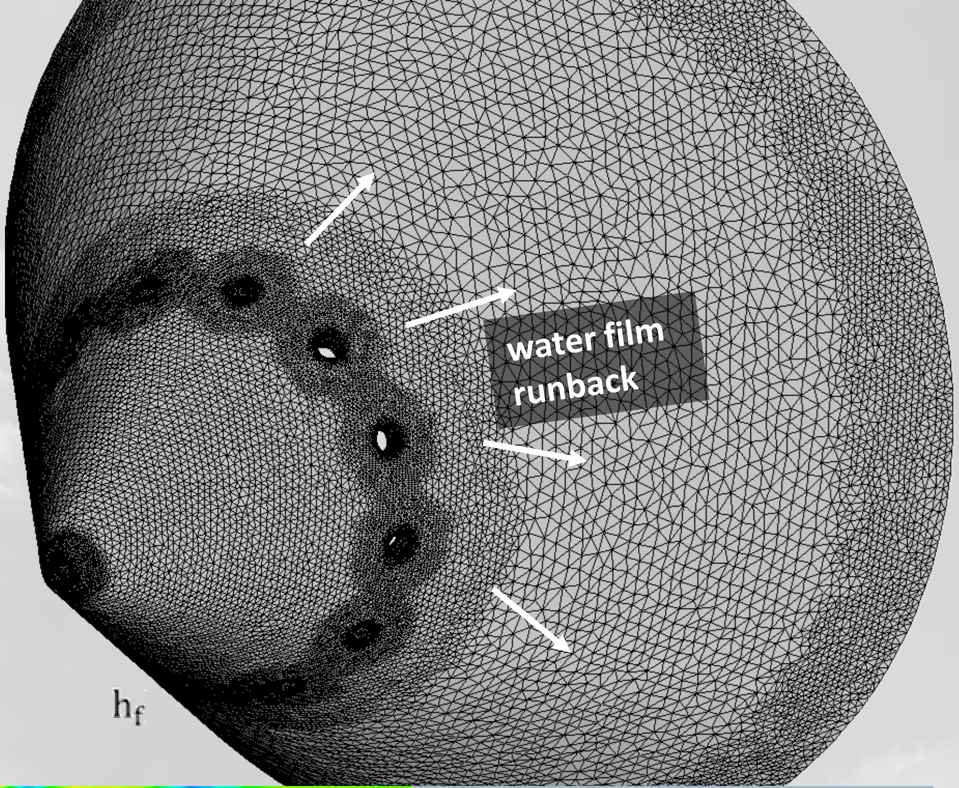


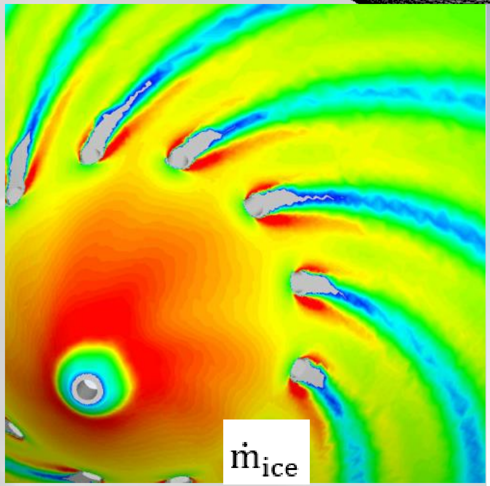
$$\underbrace{\rho_w\left[\frac{\partial h_f}{\partial t}+\nabla\cdot(\bar{u}\,h_f)\right]}_{\textbf{Transient FVM}} = \dot{m}_\beta - \dot{m}_{ice} - \dot{m}_{evap} \qquad \bar{u} = \frac{h_f}{2\mu_w}\,\bar{\tau} + f(g,\omega)$$

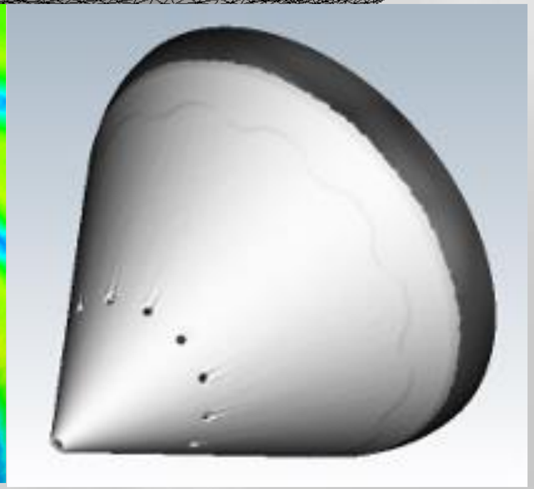
$$\rho_w\left[\frac{\partial(h_f C_w T)}{\partial t}+\nabla\cdot(\bar{u}\,h_f C_w T)\right] = \dot{Q}_\beta + \dot{Q}_{ice} - \dot{Q}_{evap} + \dot{Q}_{conv} + \dot{Q}_{cond}$$

$$\dot{Q}_{evap} = 0.7\,\boxed{\mathrm{HTC}}\,L_{evap}\,\frac{\left(P_{vap,air} - P_{vap,surf}\right)}{C_{P,air}P_{air}}$$

$$\dot{Q}_{conv} = \boxed{\mathrm{HTC}}\left(T_{rec} - T_{surf}\right) \qquad \mathrm{HTC} = \overbrace{\frac{\dot{Q}_{conv\,\boxed{RANS}}}{\left(T_{rec} - T_{wall}\right)}}^{\text{roughness, turbulence}}$$



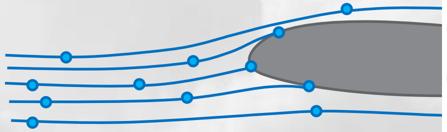


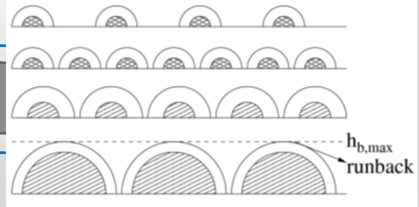


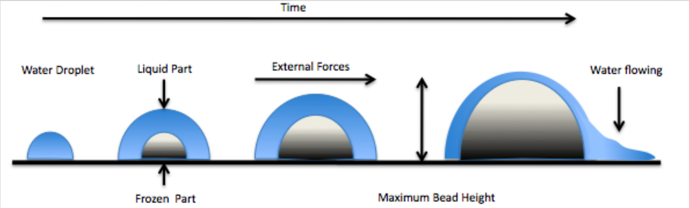
Roughness

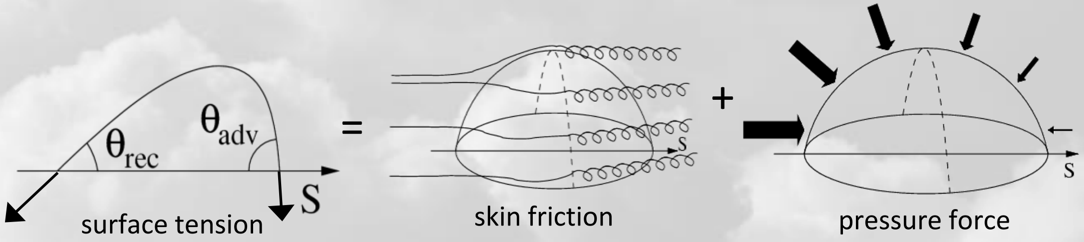
- Ice roughness setting changes the computed ice shape dramatically
- Ice roughness is not uniform, and changes with time
- Stagnation line starts out as smooth, with a lot of water film
- Roughness is higher towards the sides
  - This is a contributing factor to ice horn growth
- Beading model
  - An analytical method to compute ice roughness as part of the solution
  - Impinging droplets form small beads on the surface
  - Beads coalesce and grow
  - Growth stops when reaching a certain local bead height limit
  - Conditions for local bead height depend on local air shear and pressure gradients
- At each icing shot, new roughness distribution is used with the next iteration of the air flow solver

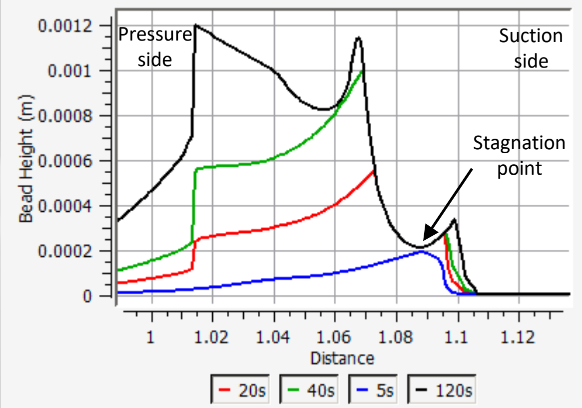
Richard Hann, NTNU UAV Icing Lab





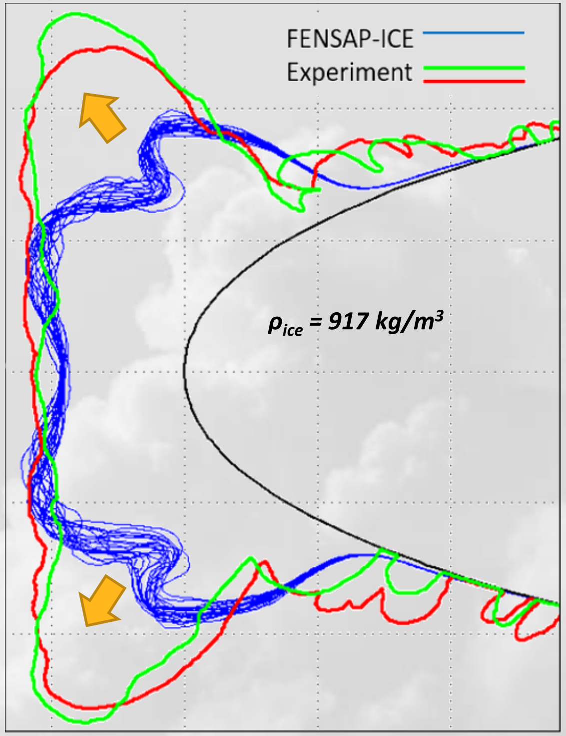


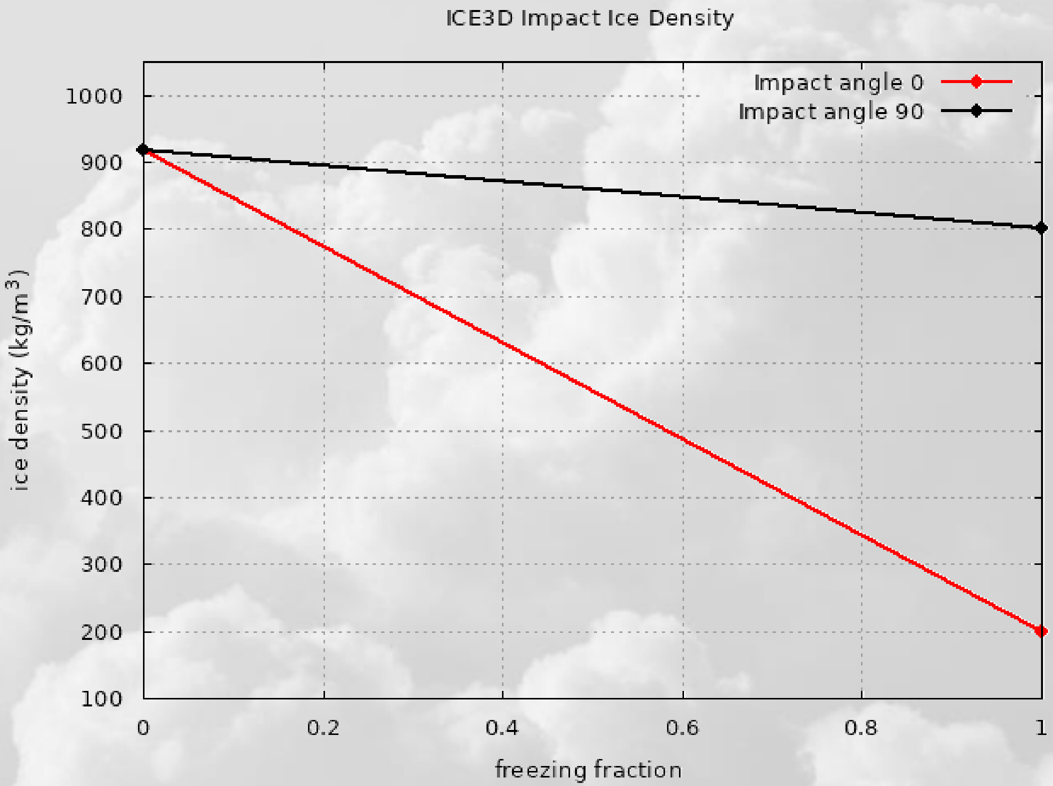


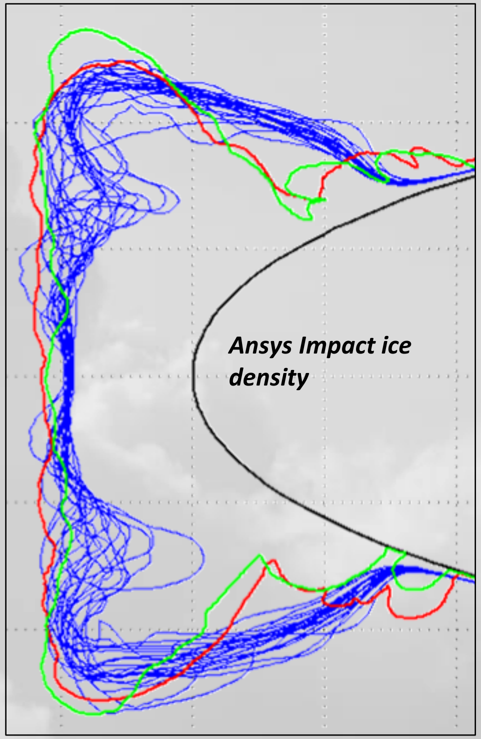


Variable ice density model development for swept wings

IPW-1
Case 362
Ts = 266K
MVD = 35µm
LWC = 0.5 g/m³
V = 103 m/s
τ = 20 min





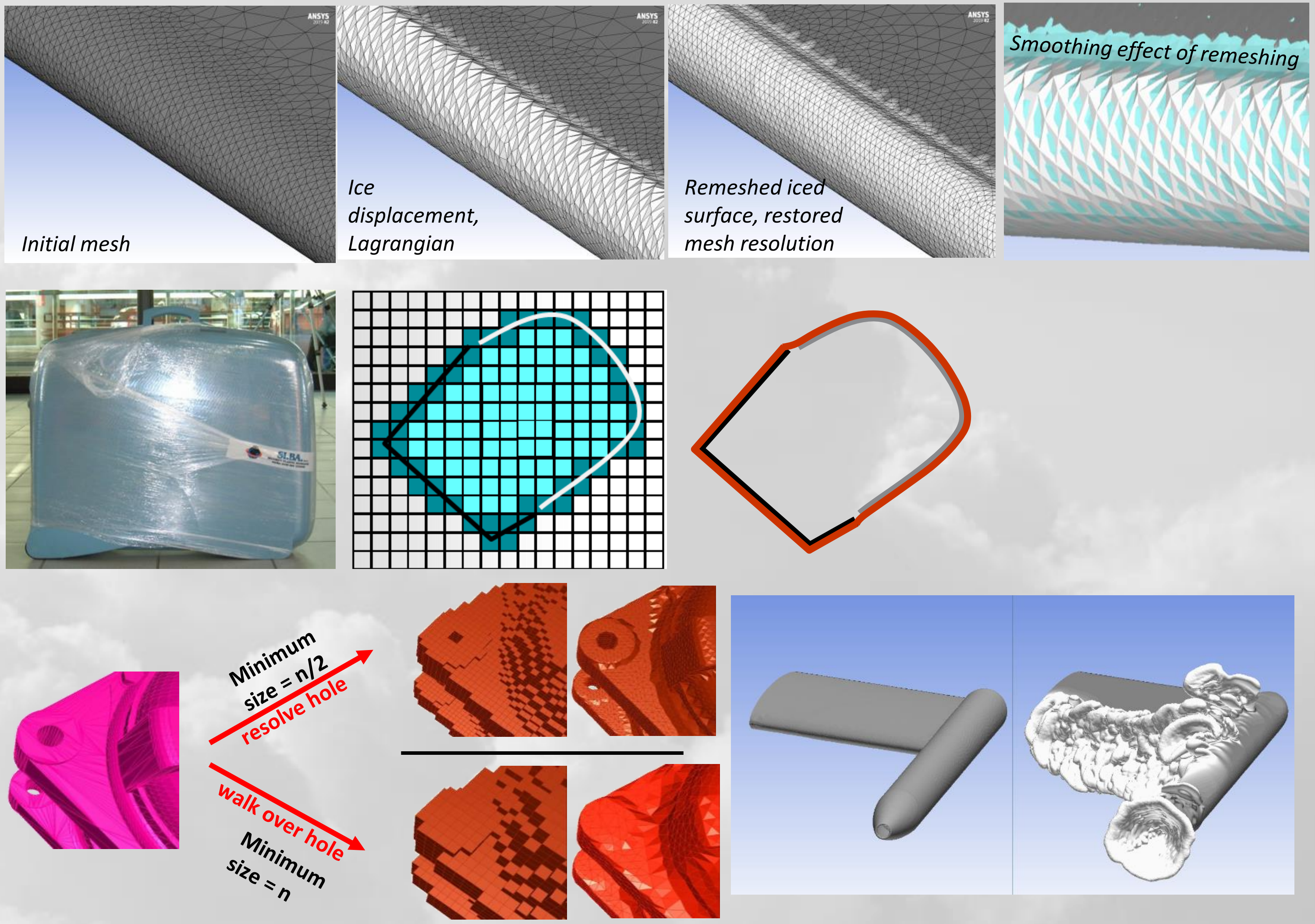


Freezing fraction = local water in-flux / freezing rate
Impact angle = Angle between droplet trajectory and surface normal
917 kg/m³ : Standard ice density, dense clear ice with FF → 0
800 kg/m³ : Rime ice density around rotating cylinders, FF = 1
~200 kg/m³ : Lowest measured ice density, rime feathers, FF = 1

Ansys

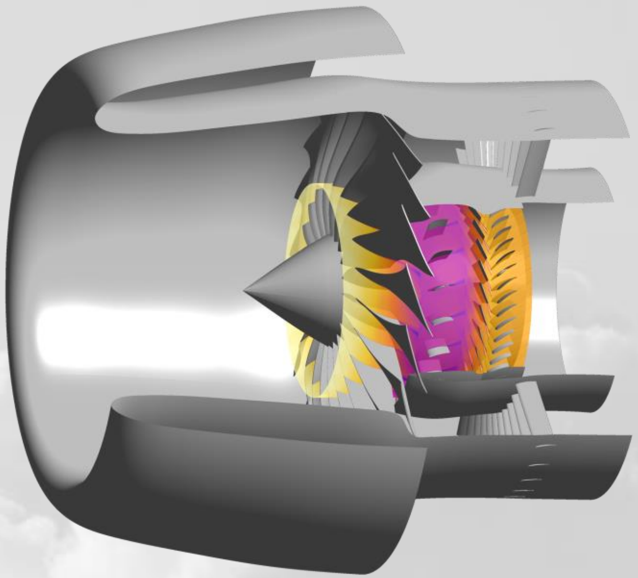
Remeshing after ice displacement

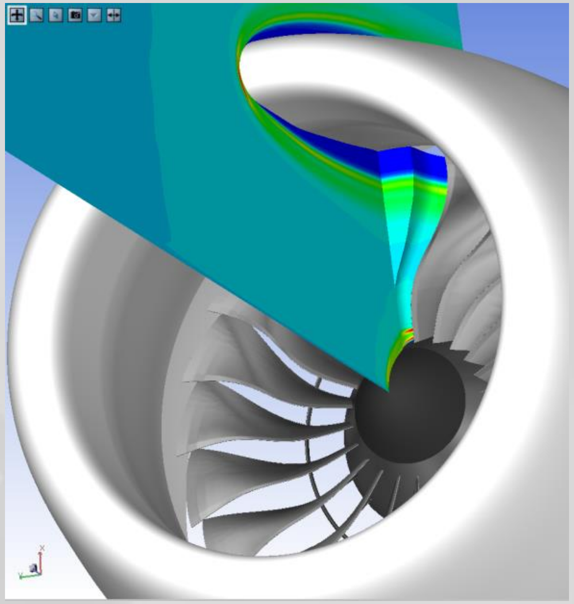
- Remeshing allows
  - Updating air and droplet solutions with intermediate ice shapes
  - Accounts for changes in collection efficiency and HTC
  - Conform surface topology to evolving ice shape
- Starting meshes can be structured, but remeshing will change them to unstructured
- Ice surface displacement initially done in a Lagrangian manner
- Shrink-wrap technique in Fluent Meshing can
  - Fill small holes
  - Blend over bridging ice shapes
  - Produce a nice isotropic surface mesh for high quality surface solutions
  - Handle mesh folds and degenerate elements produced by Lagrangian displacement
- Air, droplet, surface ice/water solutions interpolated to new mesh to restart next shot of icing calculation



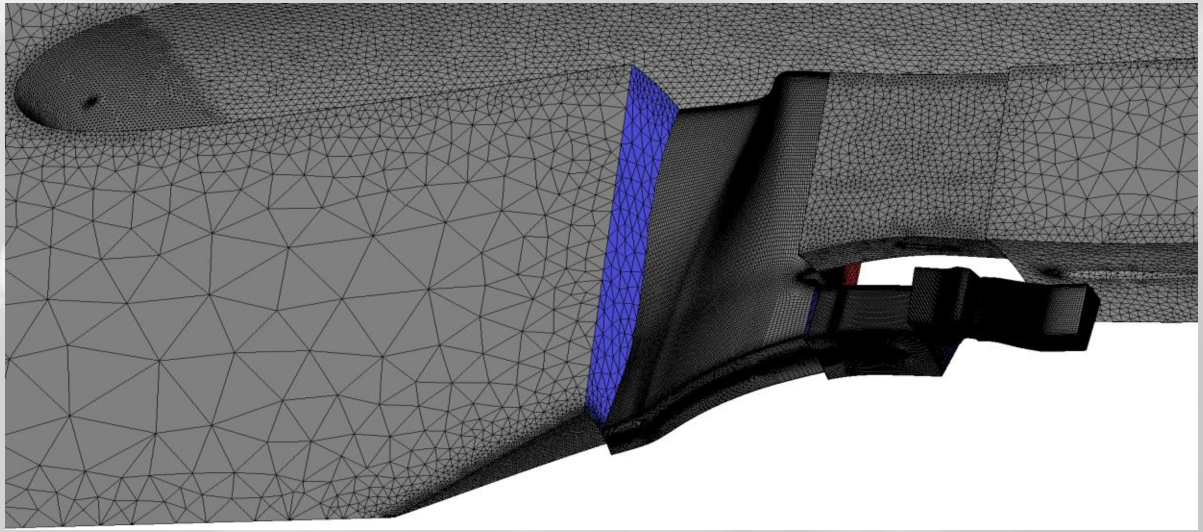
Engine icing modeling

- Steady-state, multi-row domains
- Rotationally periodic sections (full-wheel impractical)
- Mixing planes to interface components
  - Mass flux conservation requires attention
- Rotational frame formulations
- Crystals
  - Collection, bouncing and reinjection through cold components
  - Melting with particle thermal equation
  - Melt ratio control with humidity setting
  - ICI: porous ice/water mixture modeling on warm surfaces



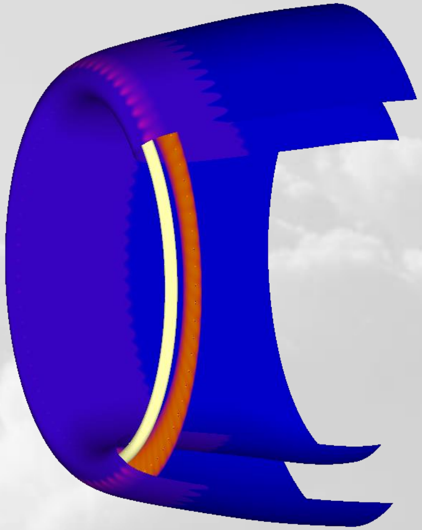


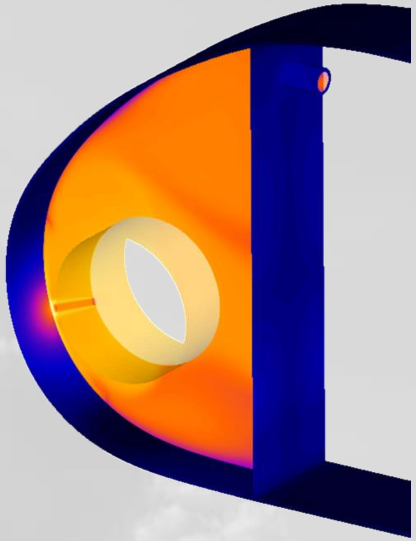
crystal reinjection from nacelle lip and spinner

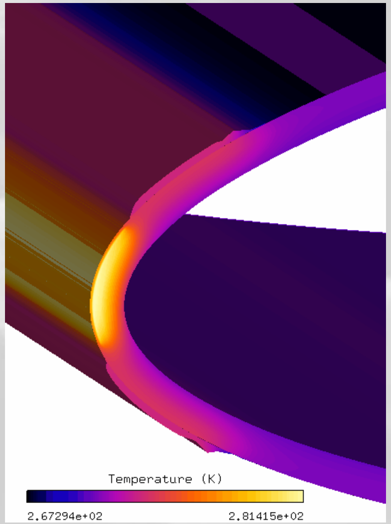


Thermal ice protection system modeling

- Piccolo tube systems for anti-icing simulations
  - 3 domains: External flow, solid, internal flow (piccolo chamber)
  - Mapped interfaces, loose coupling with HTC
  - Convergence to steady state
    - all residuals of equations reach acceptable levels
    - interface heat fluxes equalize
    - temperature distribution becomes continuous
- Electro-thermal unsteady de-icing simulations
  - 1 external, 1 solid domain
  - heat is provided by volumetric or surface type heater pads
  - Unsteady simulation, with ice layer conduction and melting
  - Heater cycle simulation
- Both simulations can compute residual ice in the unprotected regions when heating is insufficient



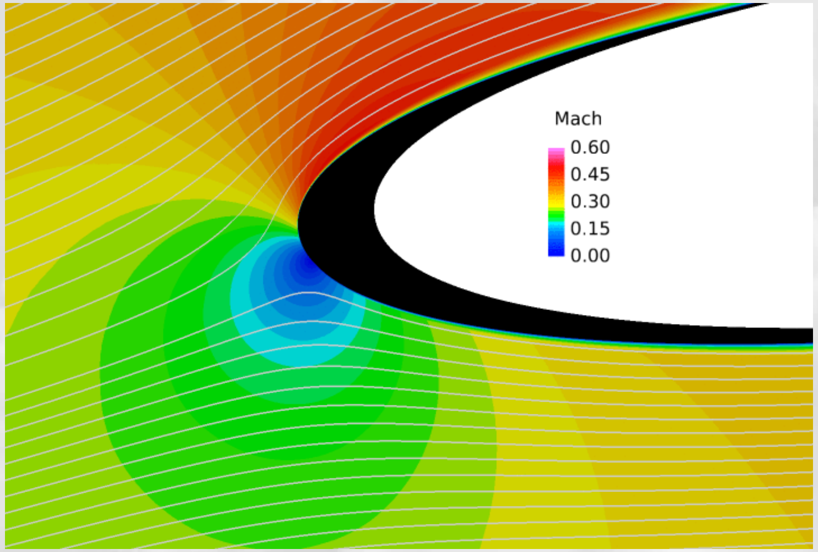


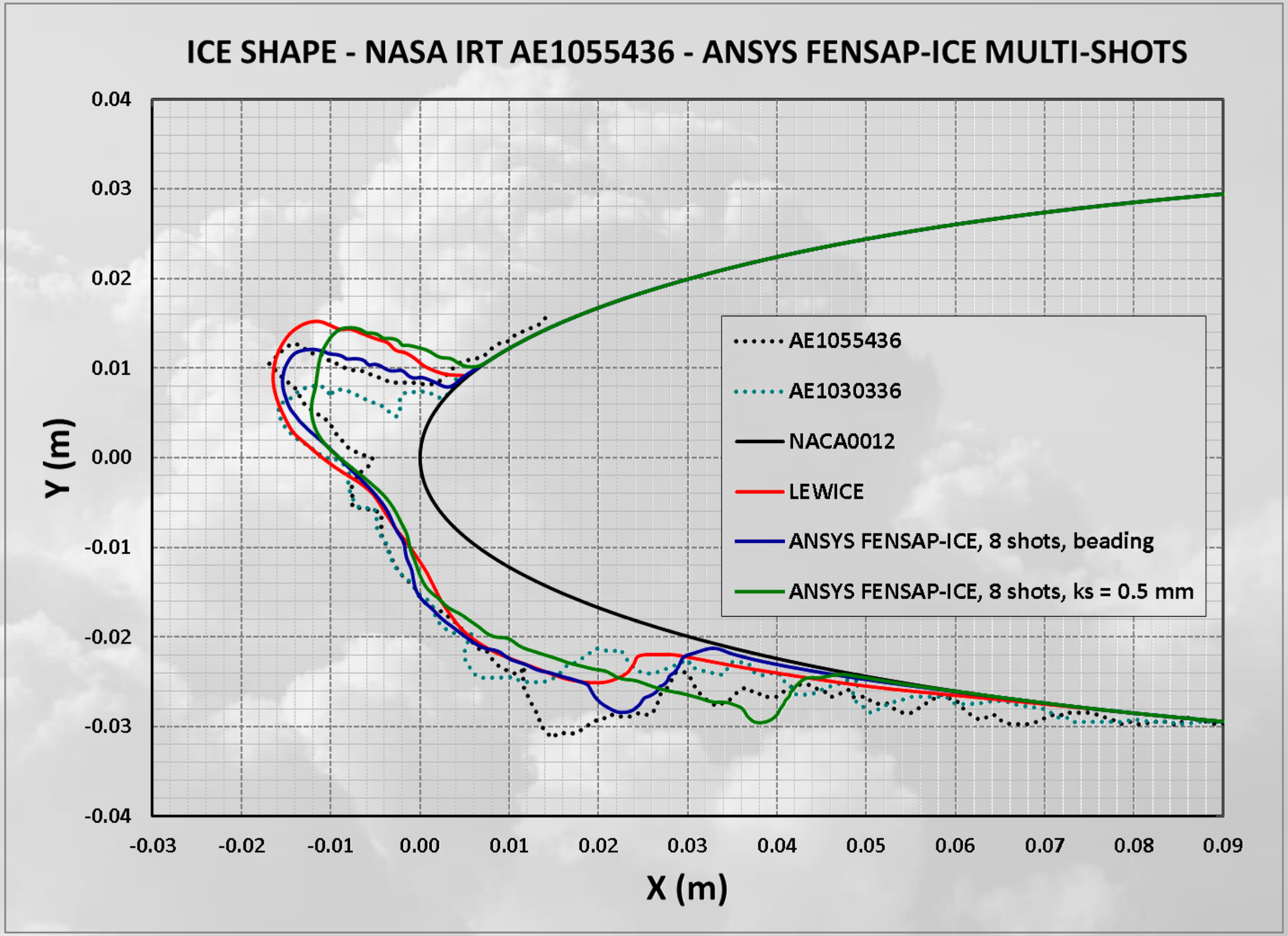


Examples and applications

2D Ice shape validations

NACA0012 (C = 0.5334 m)	
Airspeed	102.751 m/s
AoA	4.0˚
Static pressure	101,325 Pa
Static temperature	-10.808 C
LWC	1.0 g/m³
MVD	20 μm
Droplet Distribution	Monodisperse
Time	231 sec

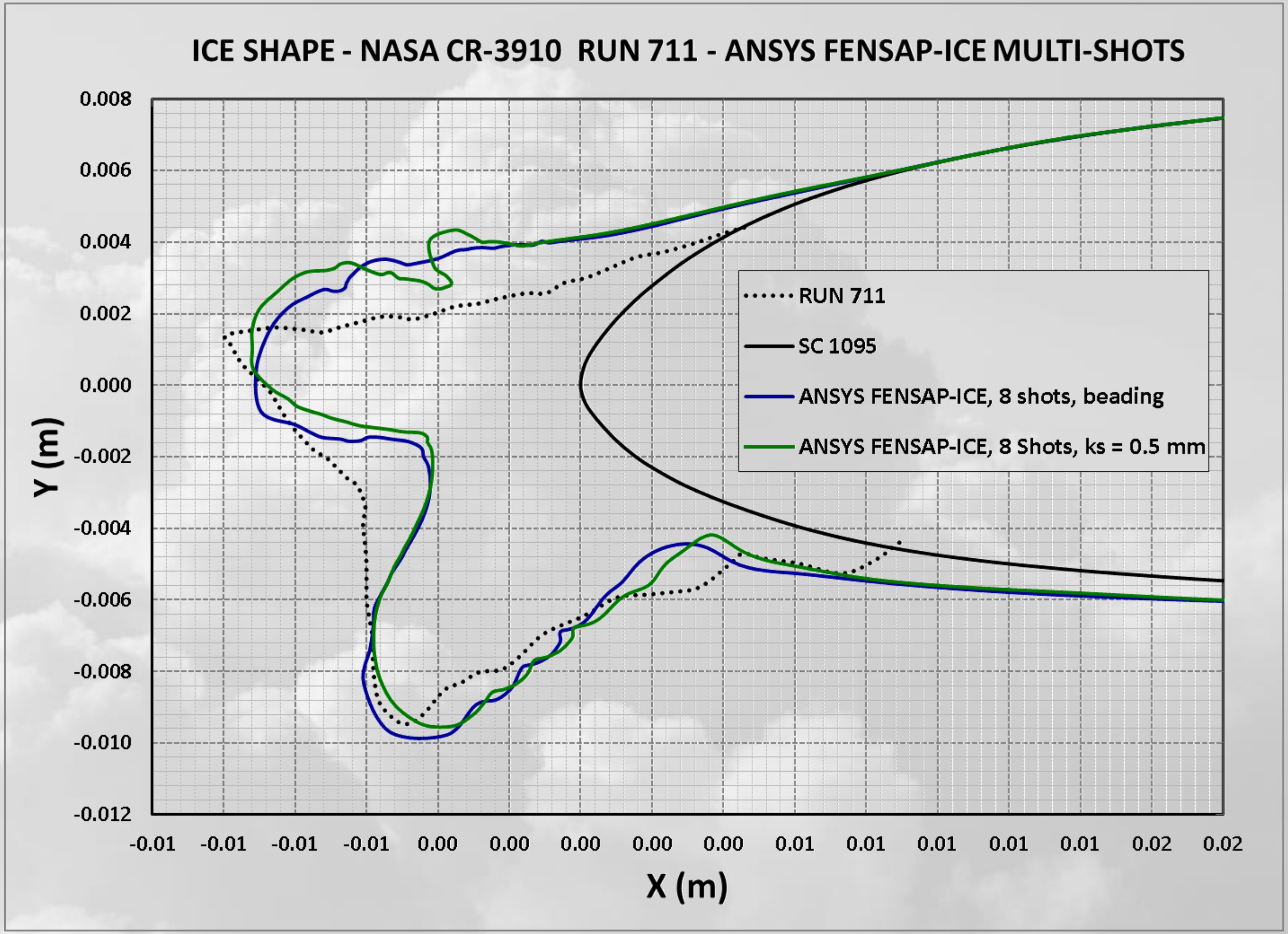




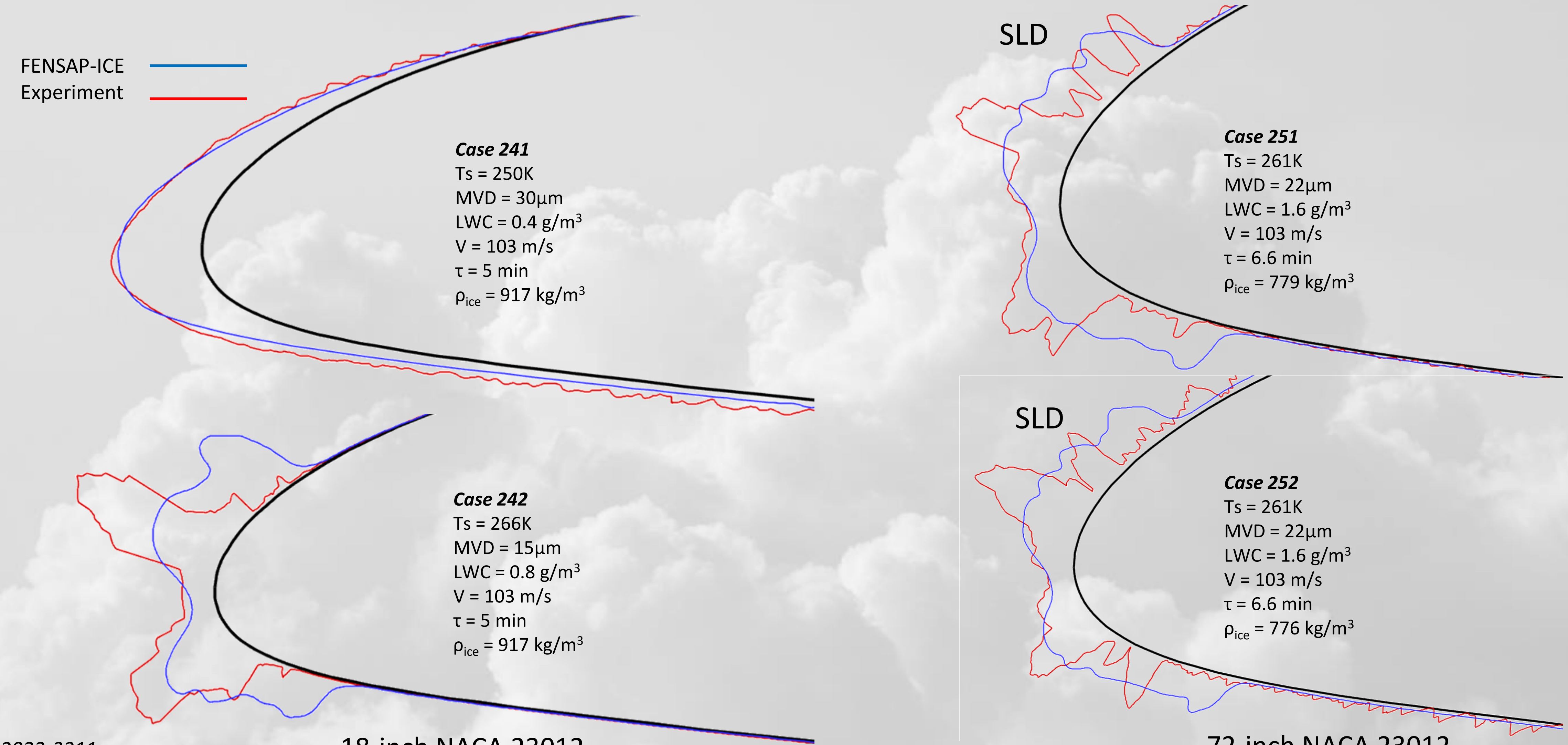
2D Ice shape validations

SC1095 (C = 0.1524 m)	
Airspeed	185.554 m/s
AoA	3.0°
Static pressure	101,325 Pa
Static temperature	-26.9 C
LWC	1.4 g/m³
MVD	20 µm
Droplet Distribution	Monodisperse
Time	45 sec

Helicopter main rotor profile,
Mach = 0.59



2D Ice shape validations, IPW-1 cases

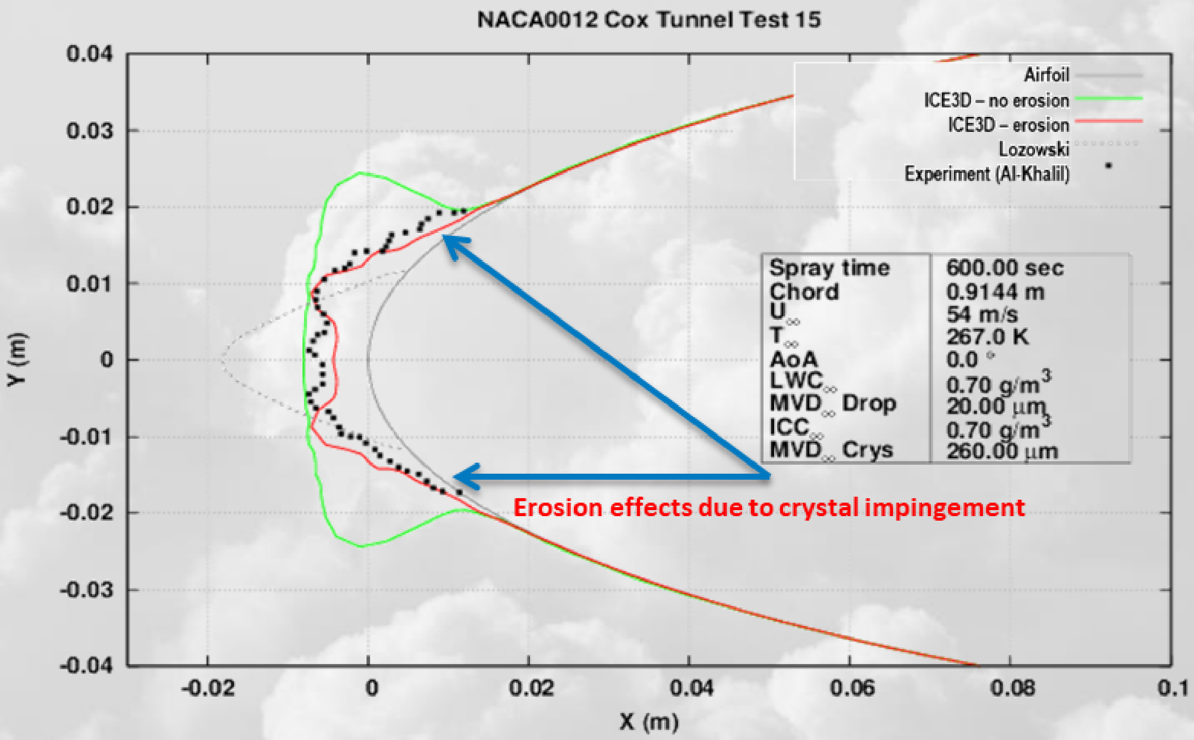


AIAA 2022-3311

18-inch NACA 23012

72-inch NACA 23012

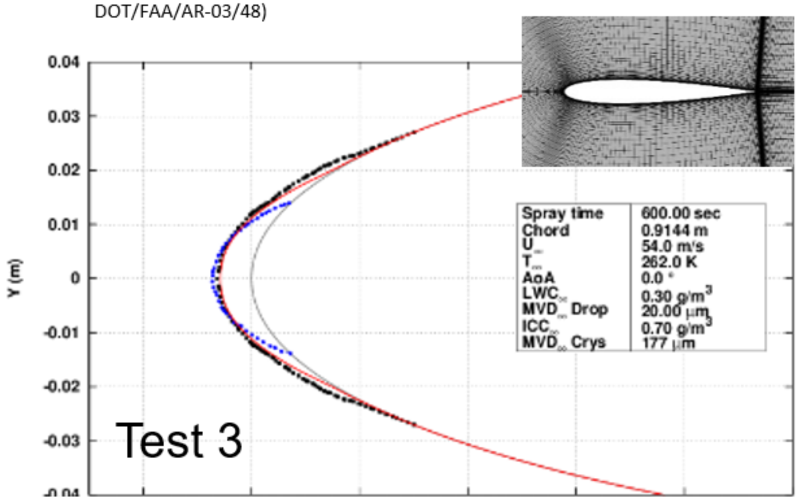
2D Ice shape validations – Droplet / Crystal mixed

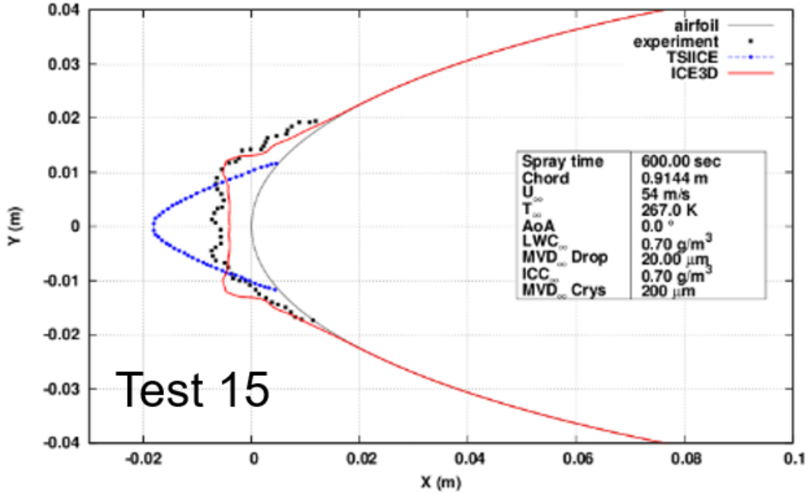


Crystal sticking, bouncing, erosion models activated

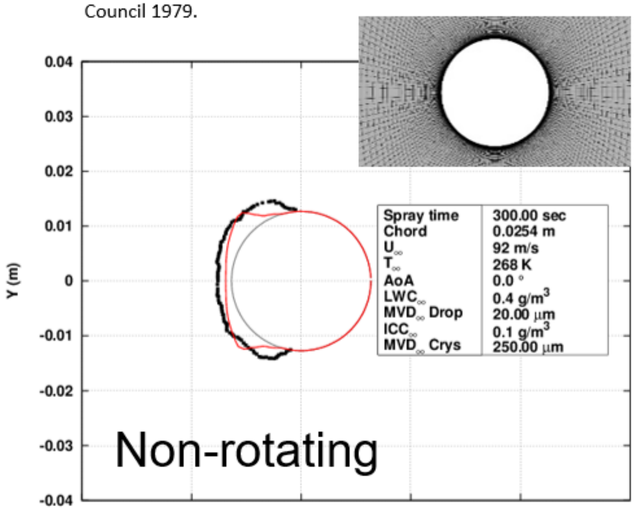
2D Ice shape validations – Droplet / Crystal mixed

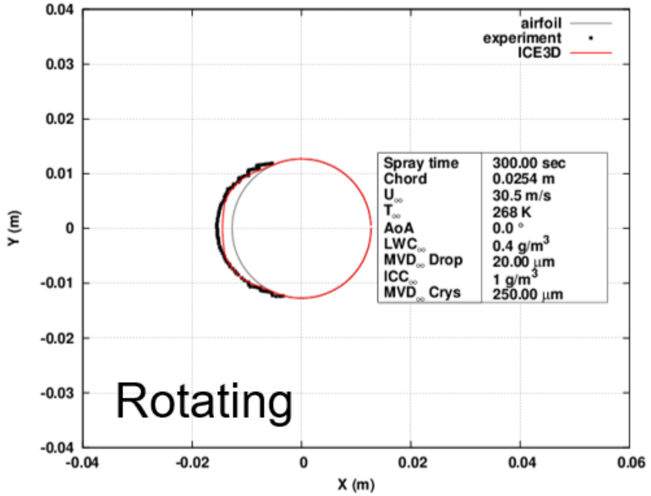
Al-Khalil, "Assessment of Effects of Mixed-Phase Icing Conditions on Thermal Ice Protection Systems", Cox & Company, Inc. , 2003 (Report No: DOT/FAA/AR-03/48)



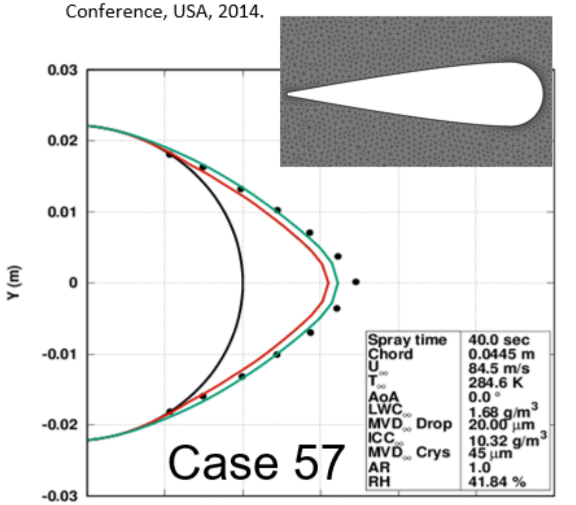


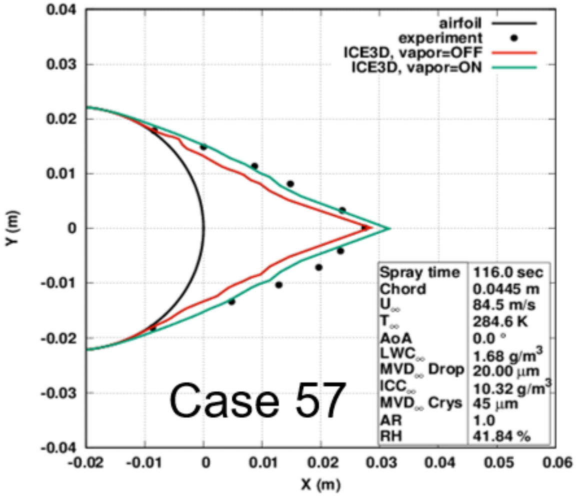
Lozowski, Stallabrass, Hearty, "The icing of an Unheated Non-Rotating Cylinder in Liquid Water Droplet – Ice Crystal Clouds", National Research Council 1979.





Currie, Fuleki and Mahallati, "Experimental Studies of Mixed-Phase Sticking Efficiency for Ice Crystal Accretion in Jet Engines," 6th AIAA Atmospheric and Space Environments Conference, USA, 2014.





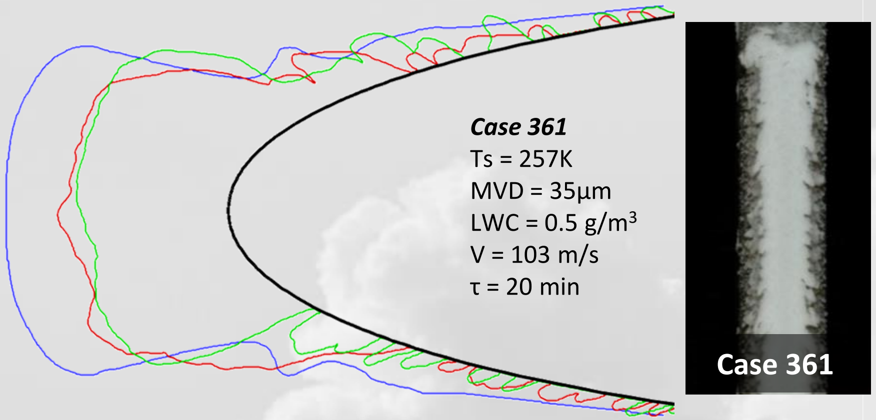
3D Ice Shape Validations: IPW-1: Swept NACA 0012 MCCS

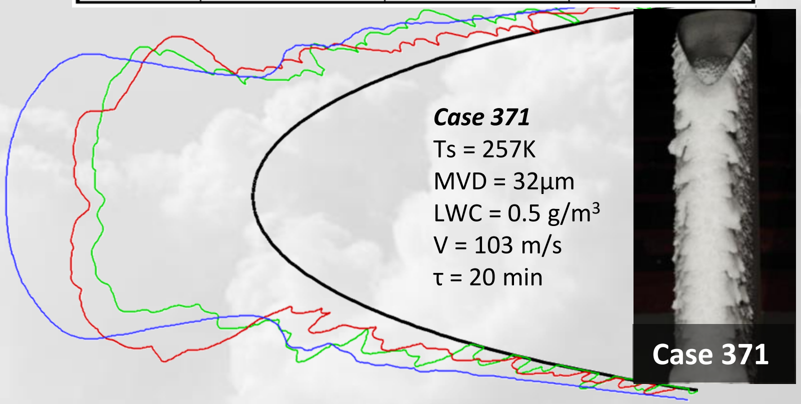
MCCS: Maximum combined cross-section

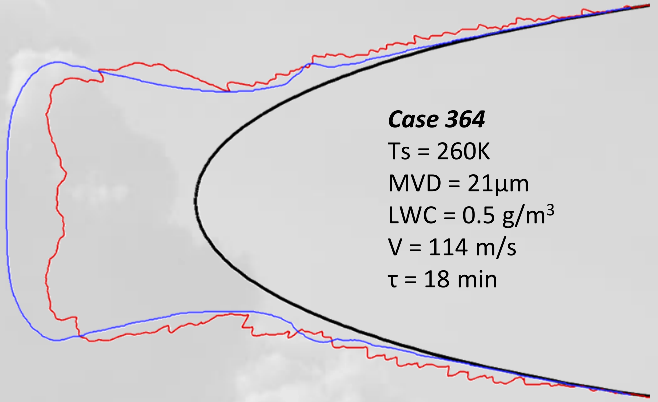
FENSAP-ICE MCCS, impact ice density model

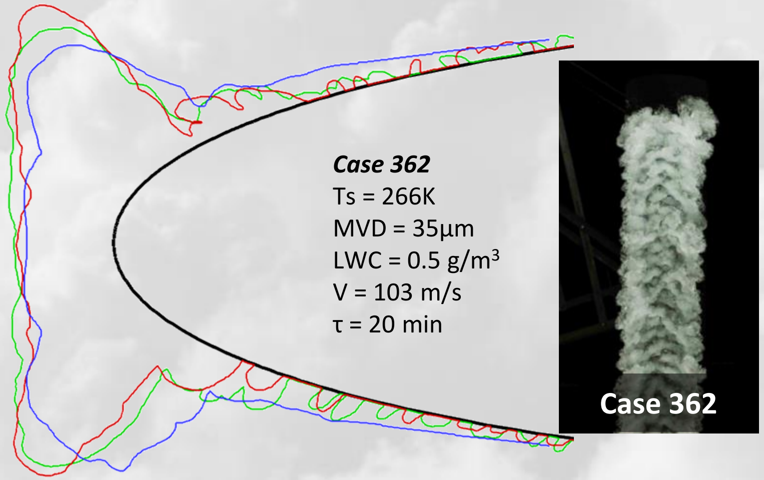
Experiments

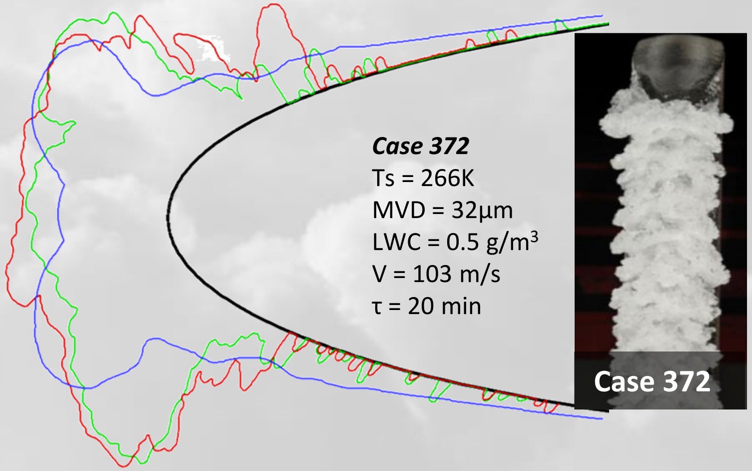
Case	Experimental Ice Mass (Kg/m)	Corrected Ice Mass (Kg/m)	Ansys Ice Mass (Kg/m)
361	1.63	1.90	2.20
362	1.43	1.62	1.70
363	1.56	N/A	1.42
364	1.38	N/A	1.44
371	1.48	1.58	1.62
372	1.45	1.60	1.50

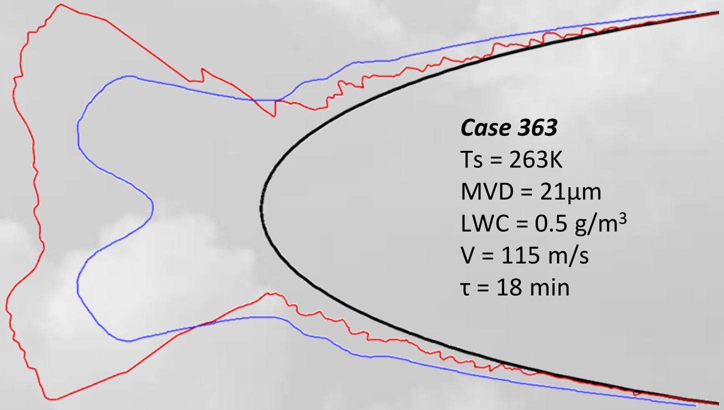












AIAA 2022-3311

30° swept NACA 0012

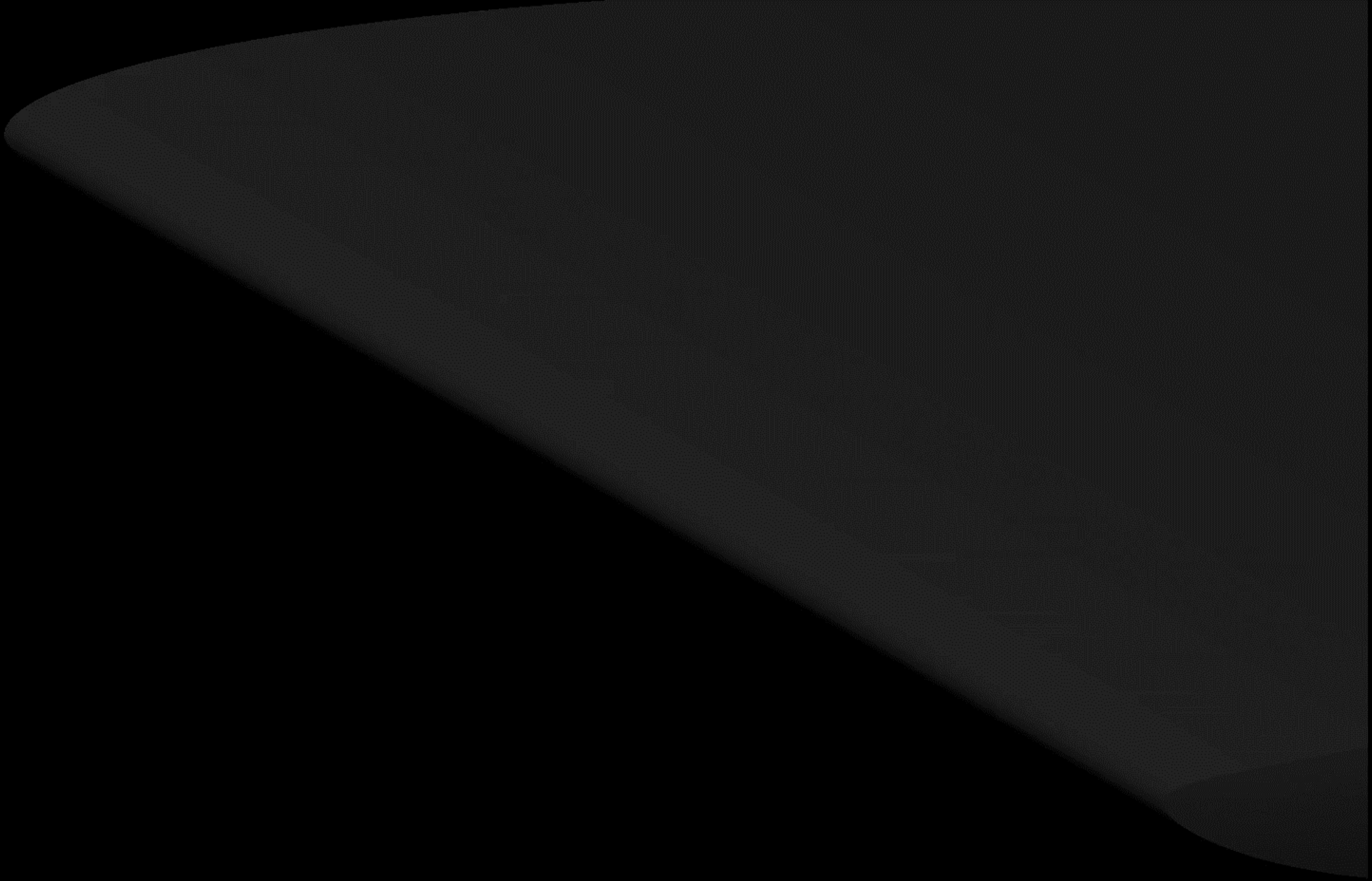
45° swept NACA 0012

30° swept NACA 0012

Ansys

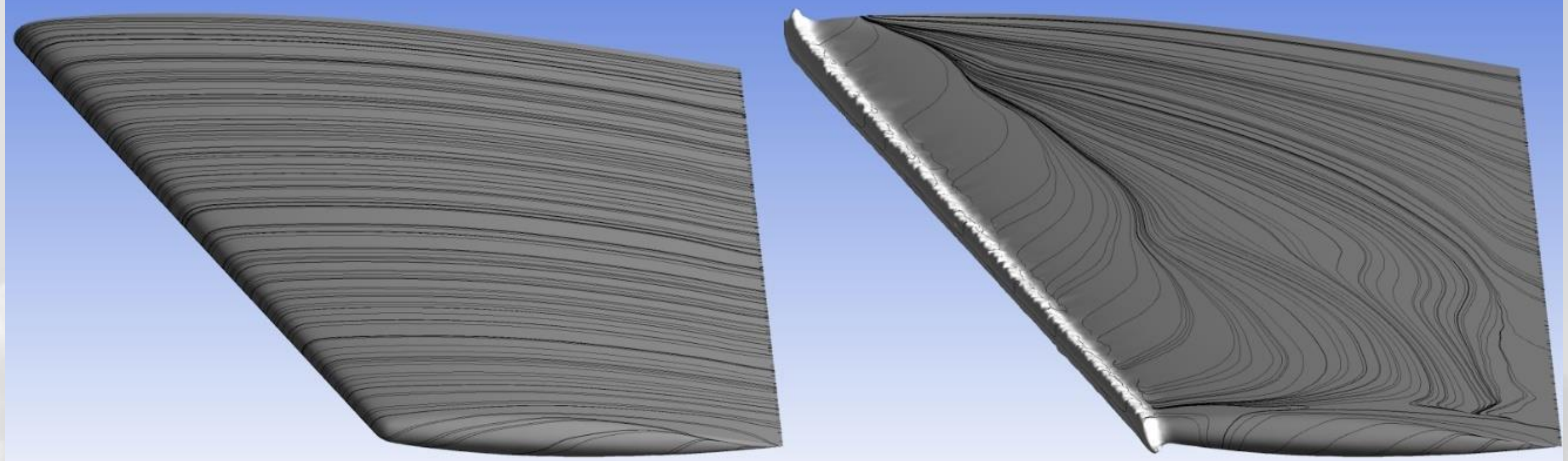
IPW-1
Case 362
Ts = 266K
MVD = 35μm
LWC = 0.5 g/m^3
V = 103 m/s
τ = 20 min

40-shot simulation

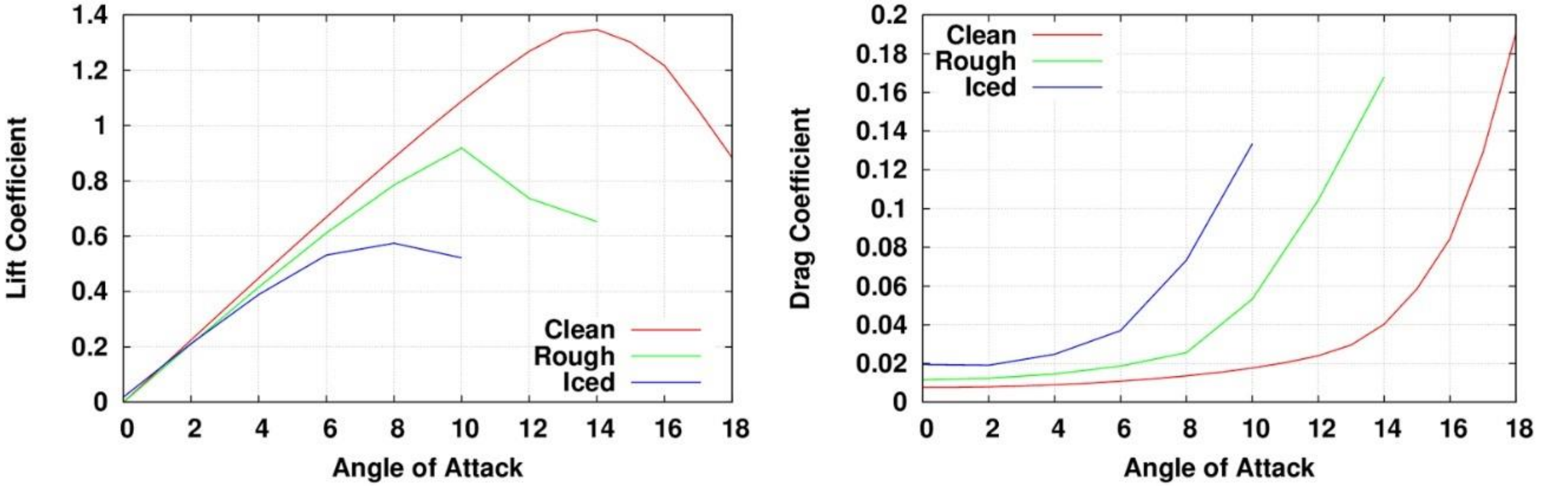


AIAA 2022-3311

Ice-induced aerodynamic penalty for a H-tail

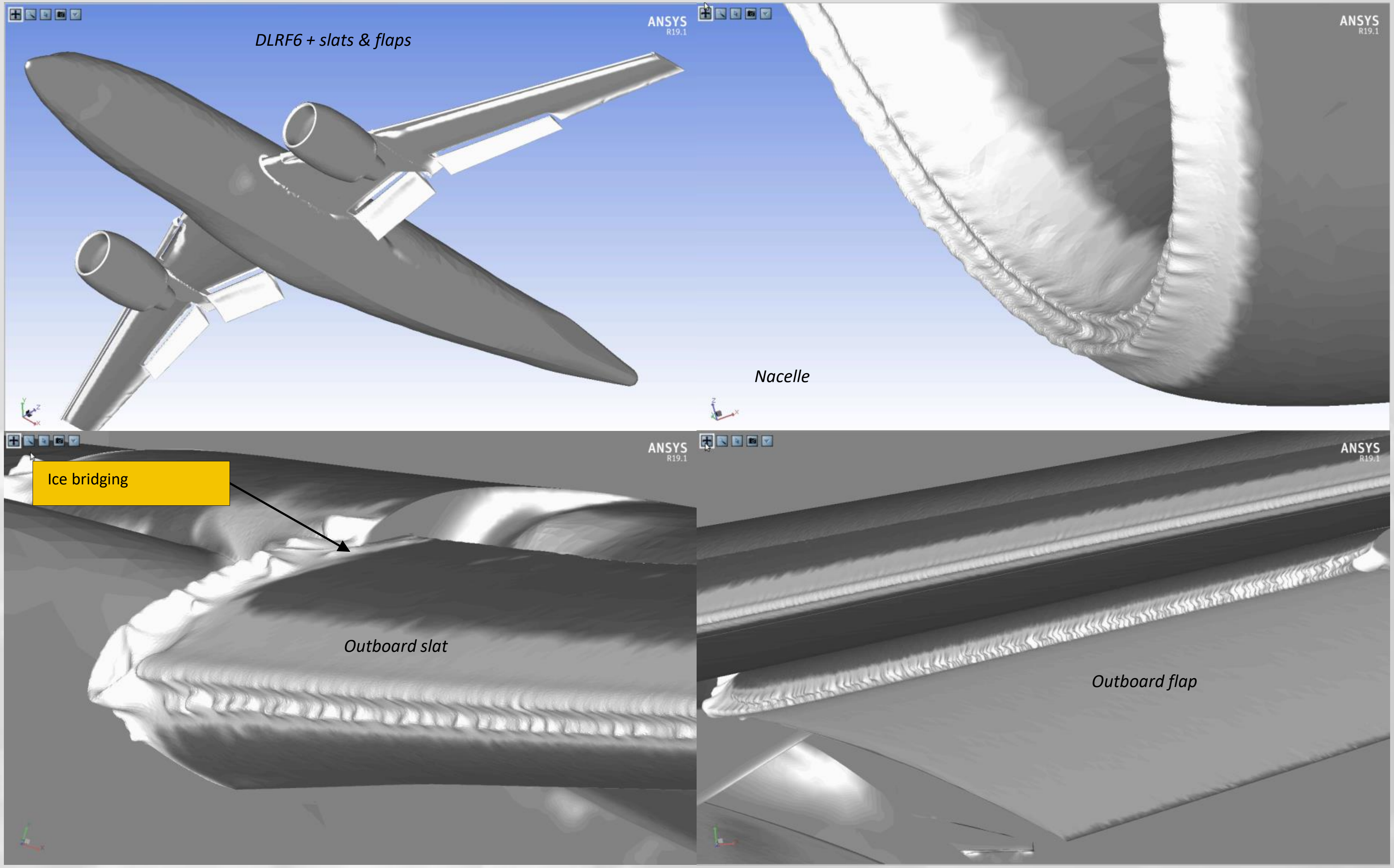


Surface oil lines on a horizontal stabilizer: Clean (left) and after 15 minutes of icing (right)



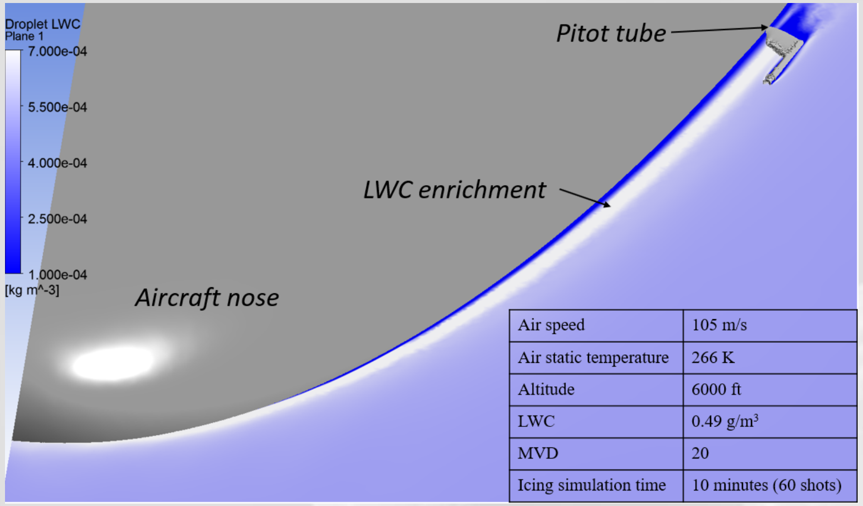
Ansys

Full aircraft icing in 45-minute holding conditions

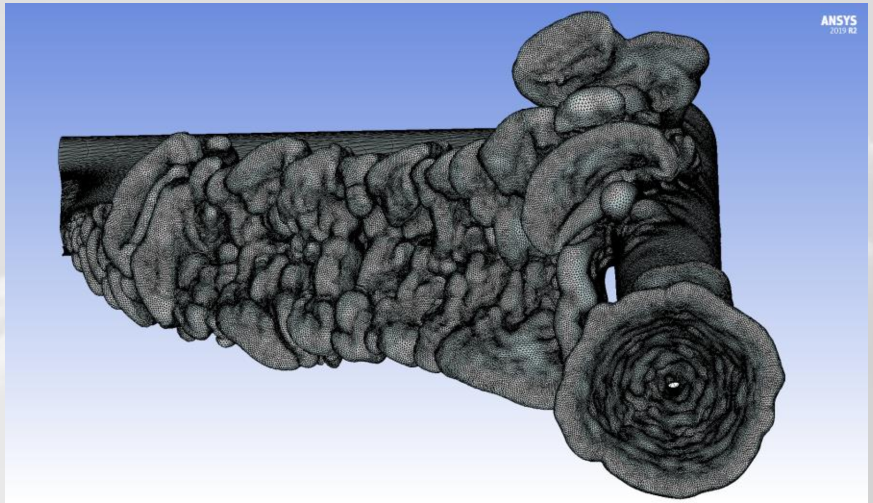


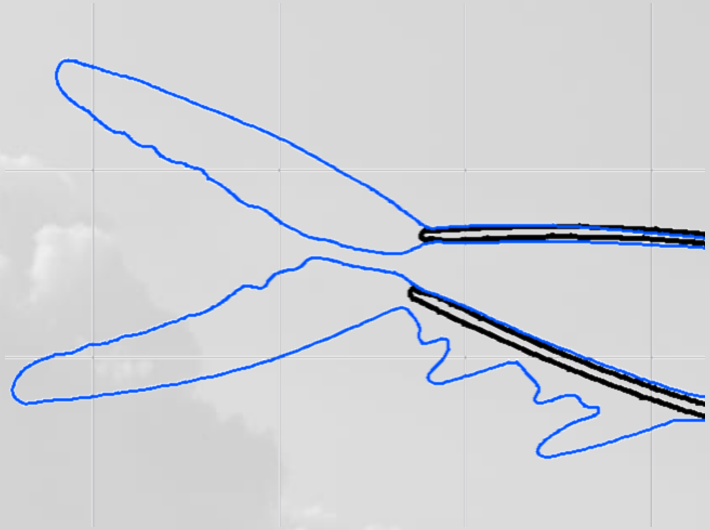
SAE 2019-01-1956

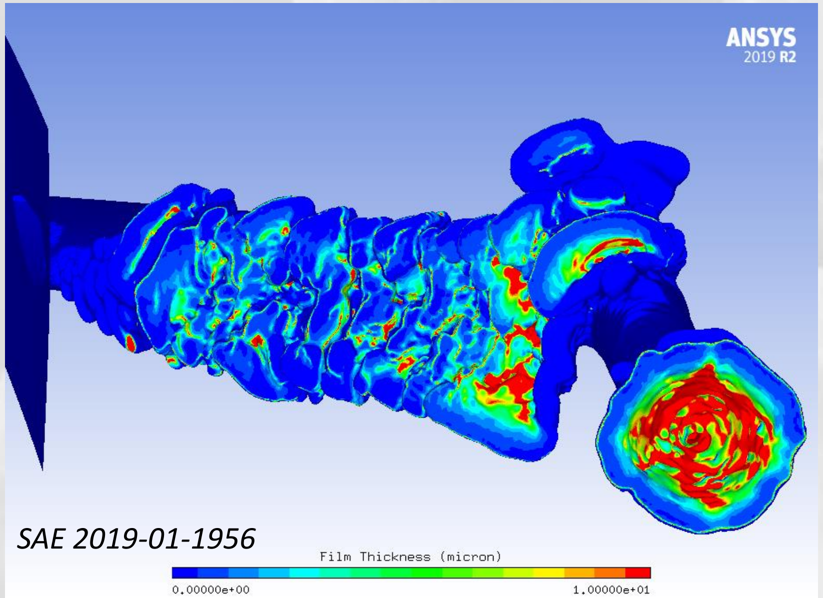
Pitot tube icing as installed on aircraft

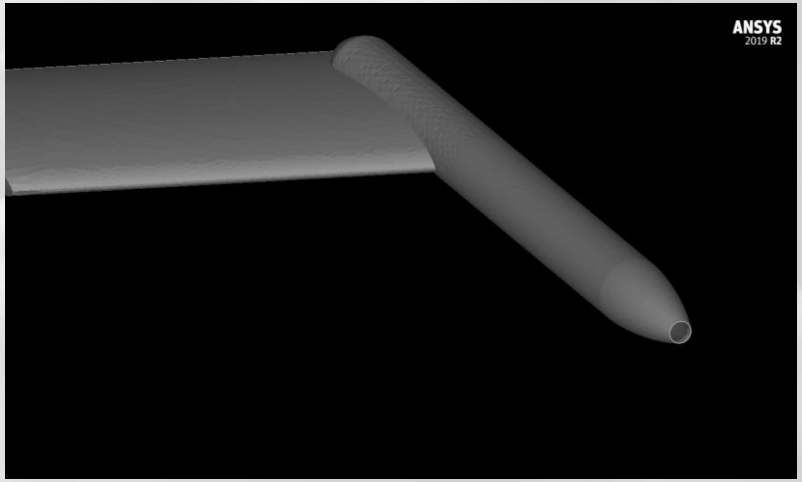


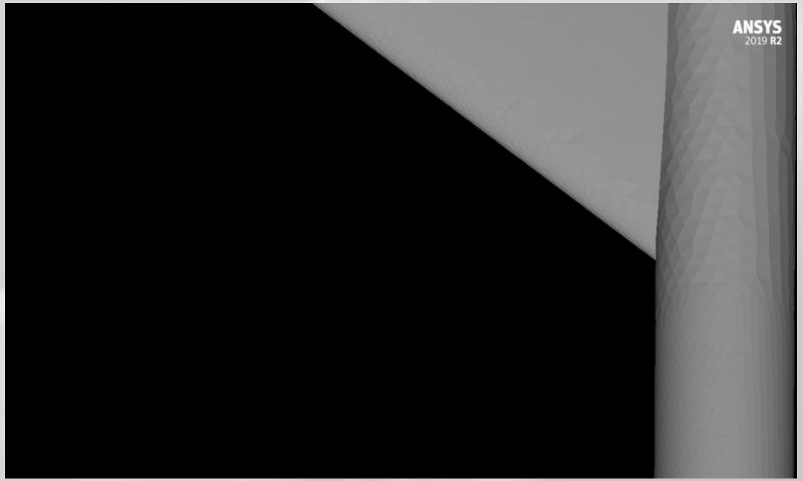
Air speed	105 m/s
Air static temperature	266 K
Altitude	6000 ft
LWC	0.49 g/m^3
MVD	20
Icing simulation time	10 minutes (60 shots)



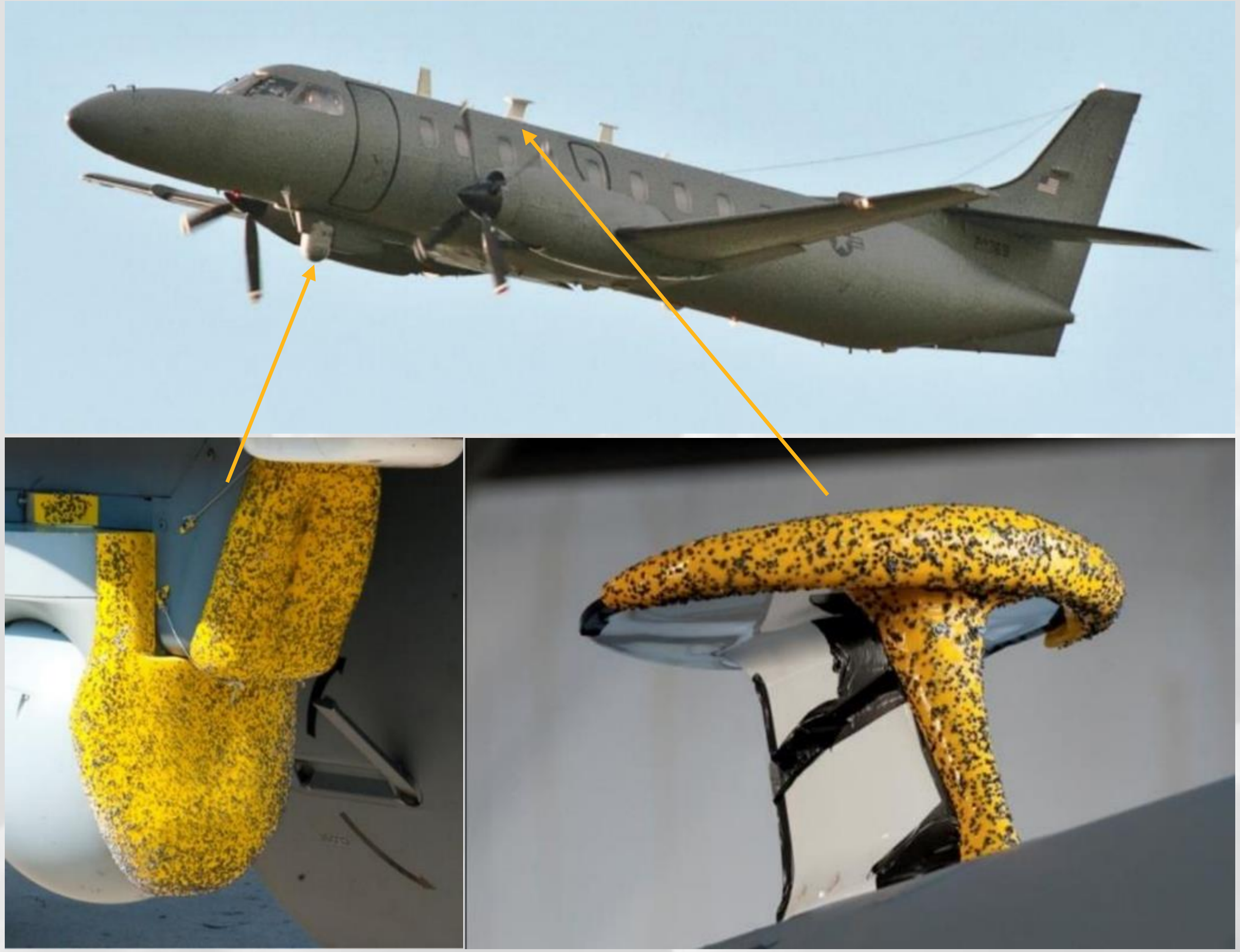


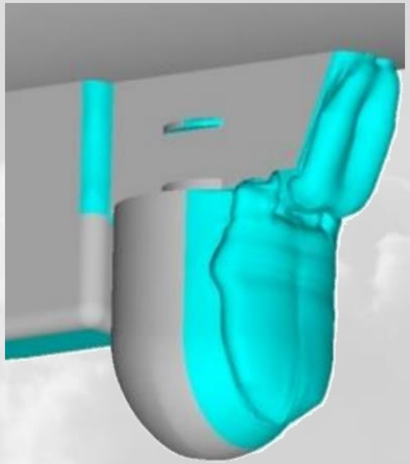


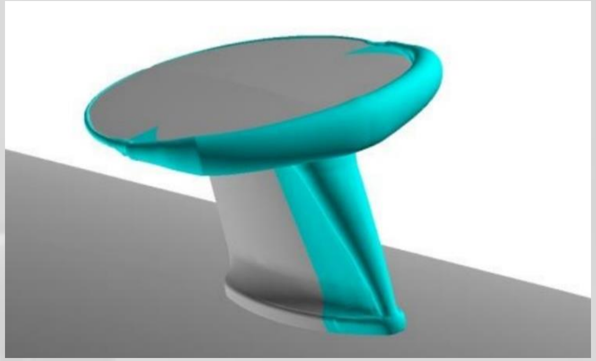


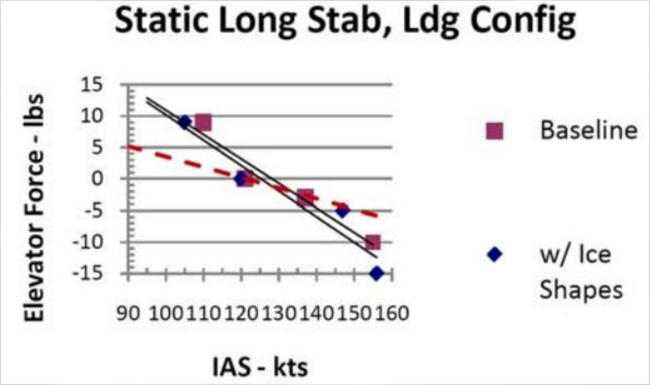


Aid-to-certification





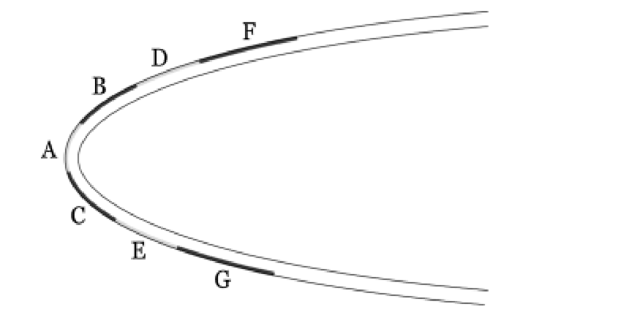




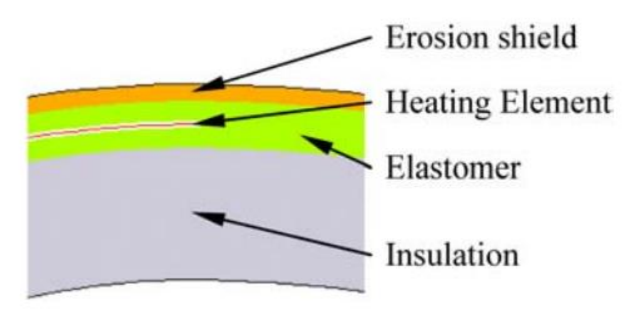
One RC-26B aircraft originally certificated for flight into known icing conditions was retrofitted with a FLIR sensor and a SATCOM antenna, which prohibited its flight into icing environments. Ice shapes predicted using FENSAP-ICE were 3D-printed and installed on the airplane to show that ice build-up on these new components did not change the control characteristics of the aircraft, and the flight into known icing restriction was removed. – *SAE 2011-38-0033*

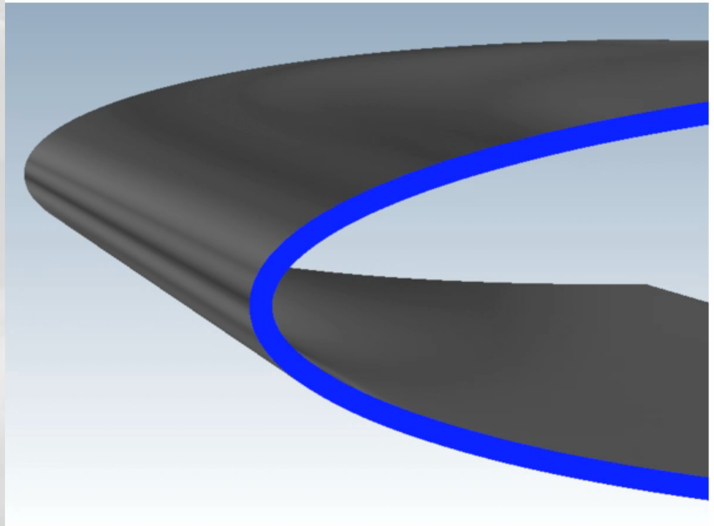
Electro-thermal de-icing of an airfoil

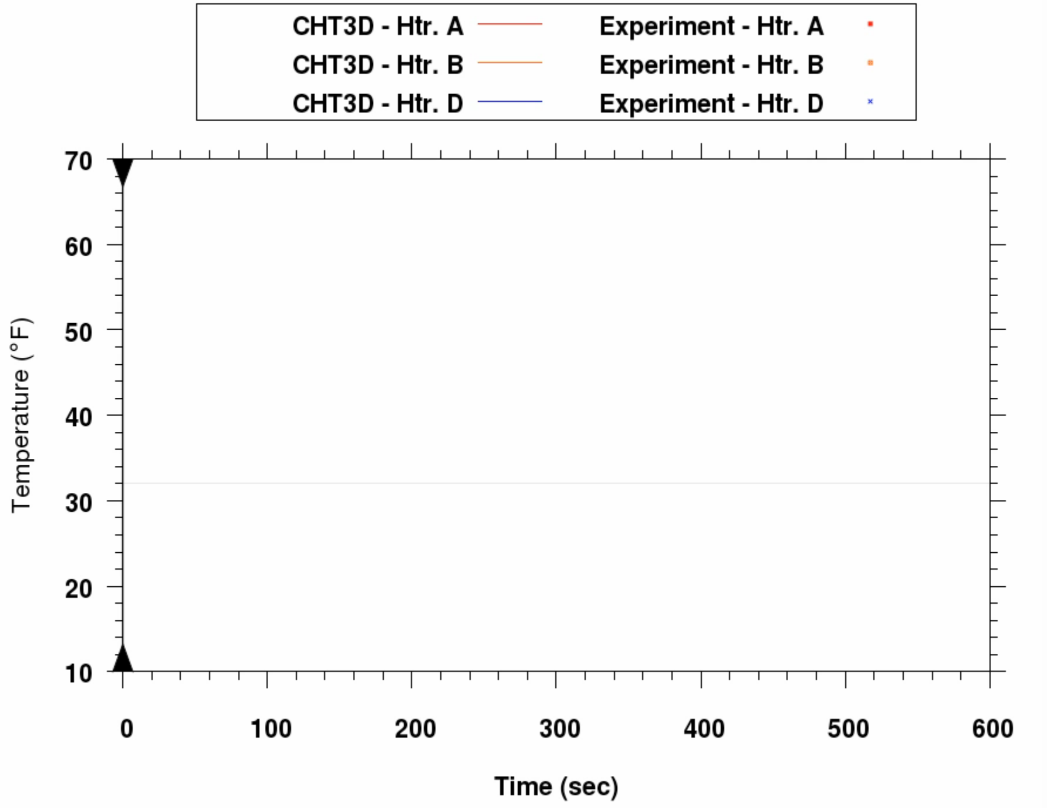
Stagnation temperature	20°F
Velocity	100 mph
Liquid water content	0.78 g/m^3
Mean volume diameter	20 μm
Heater A power density	5 $W/in.^2$
Heaters B-C power density	10 $W/in.^2$
Heaters D-G power density	8 $W/in.^2$



IPS heater pad arrangement on the NACA 0012 airfoil.

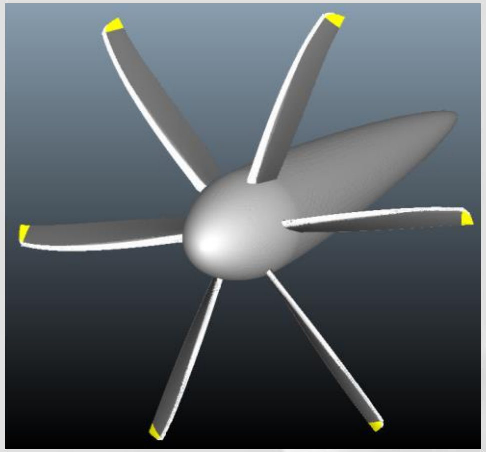


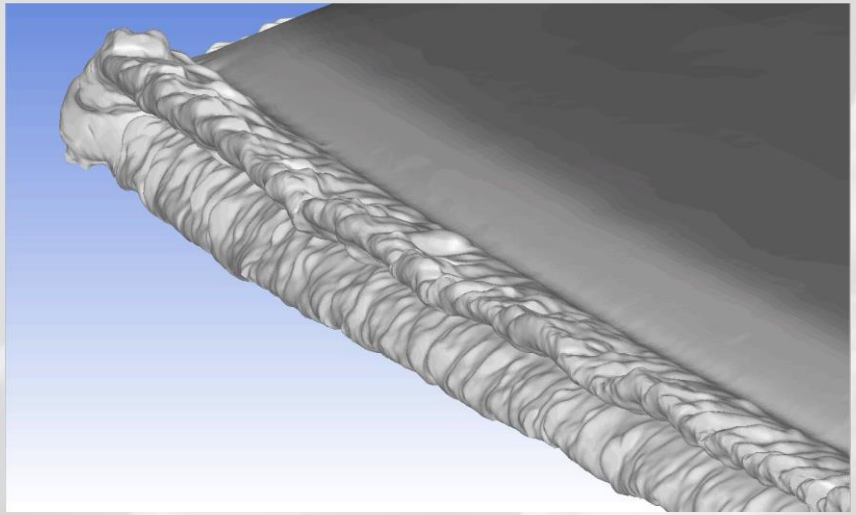


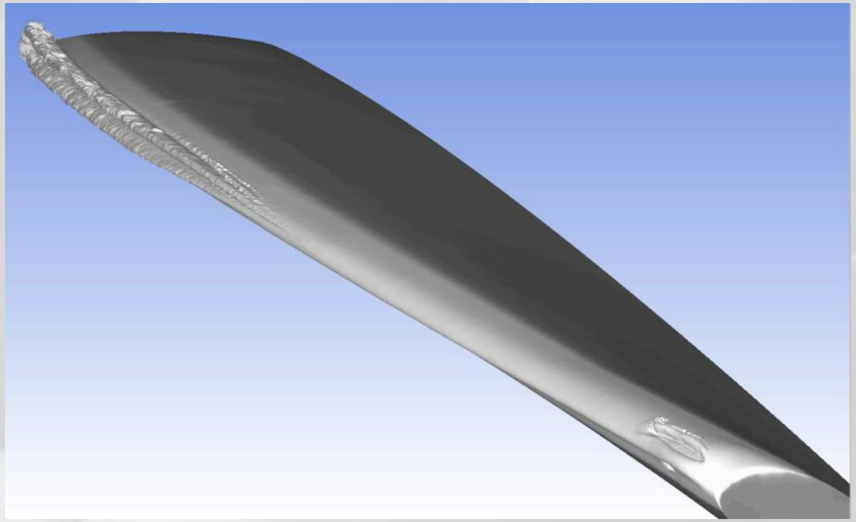


Refs: JOURNAL OF AIRCRAFT Vol. 49, No. 4, July–August 2012, FENSAP-ICE: Unsteady Conjugate Heat Transfer Simulation of Electrothermal De-Icing , "Validation of NASAThermal Ice Protection Computer Codes. Part 1: Program Overview," AIAA Paper 97-0049, 1997

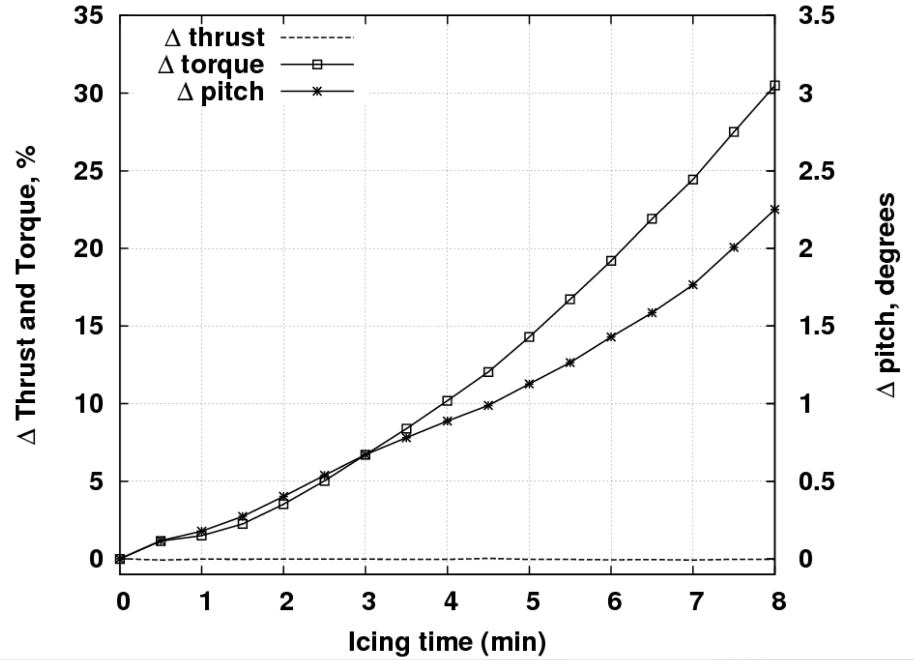
Propeller trimming during icing







Air speed	190 kts
Air static temperature	250 K
Angle of attack	2.5
Altitude	6000 ft
Droplet MVD	20 microns
LWC	0.2 g/m^3
Icing time	10 minutes (10 shots)
Propeller speed	850 rpm

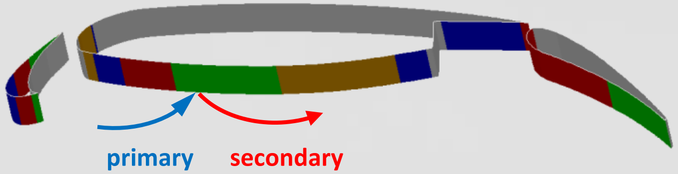


Propeller iteratively trimmed after each icing shot to restore thrust

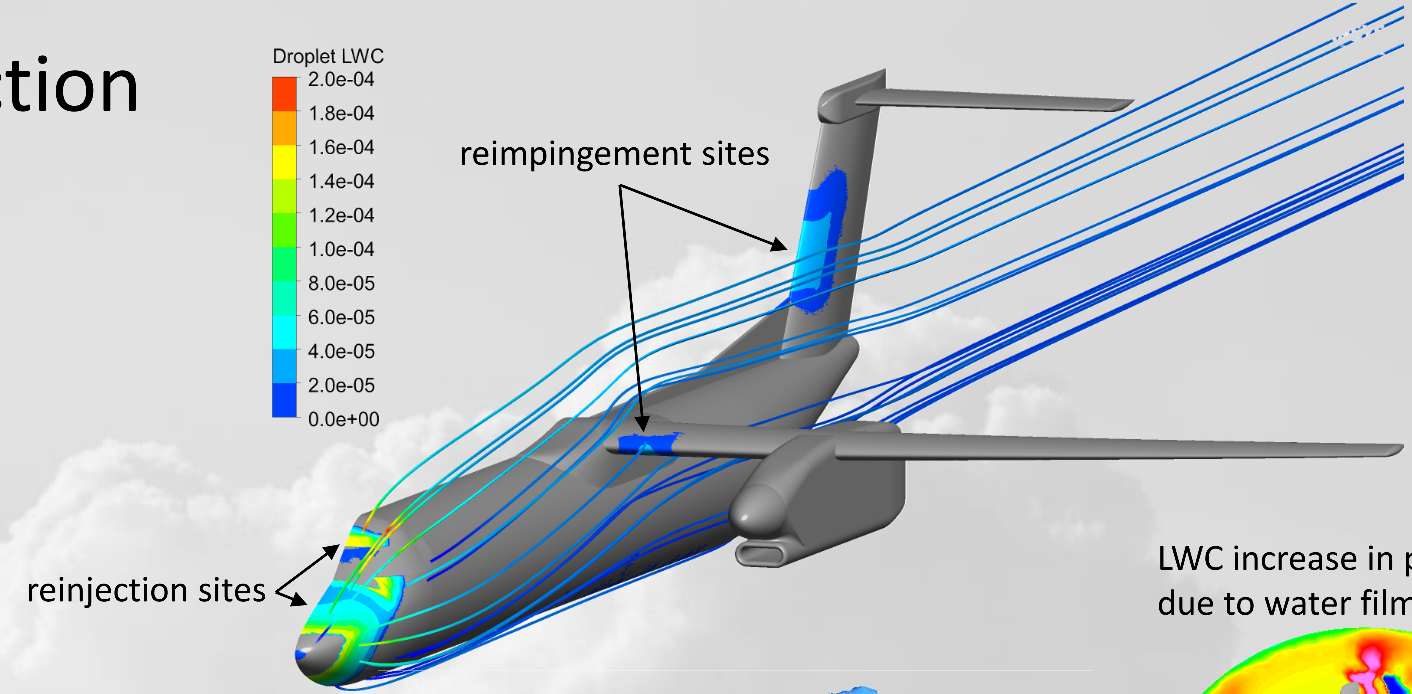
In 10-minute App-C icing, torque increased by 30%

SAE 2019-01-1954

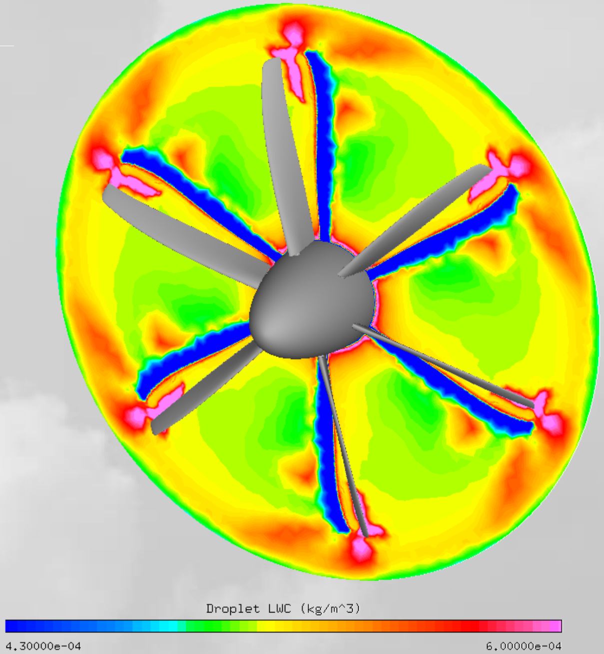
SLD and film Reinjection



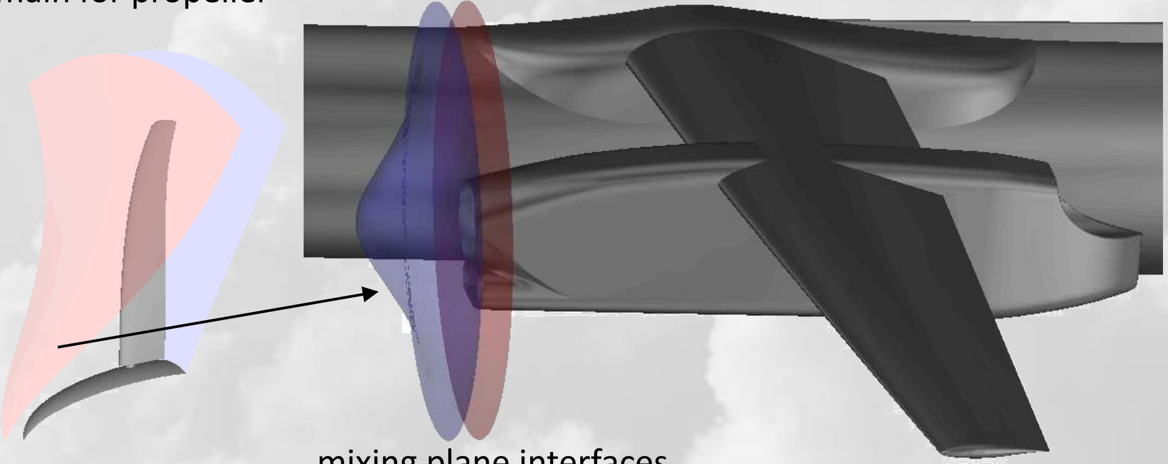
SLDs can bounce and reinject
These “secondary” tracks are released section by section



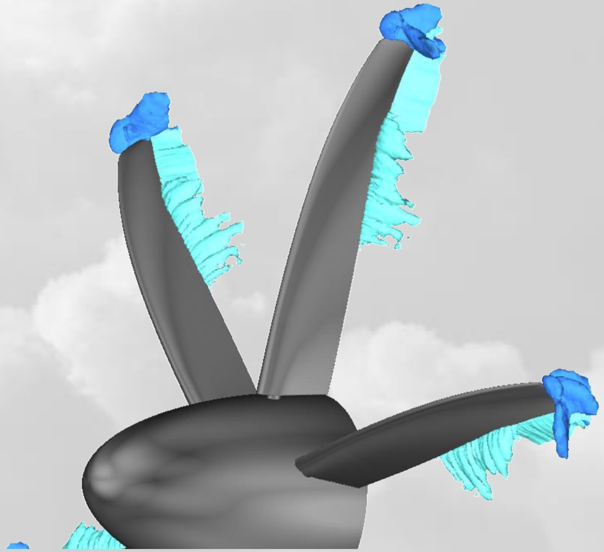
LWC increase in prop wake due to water film injection



rotationally periodic domain for propeller



mixing plane interfaces with full-wheel mesh

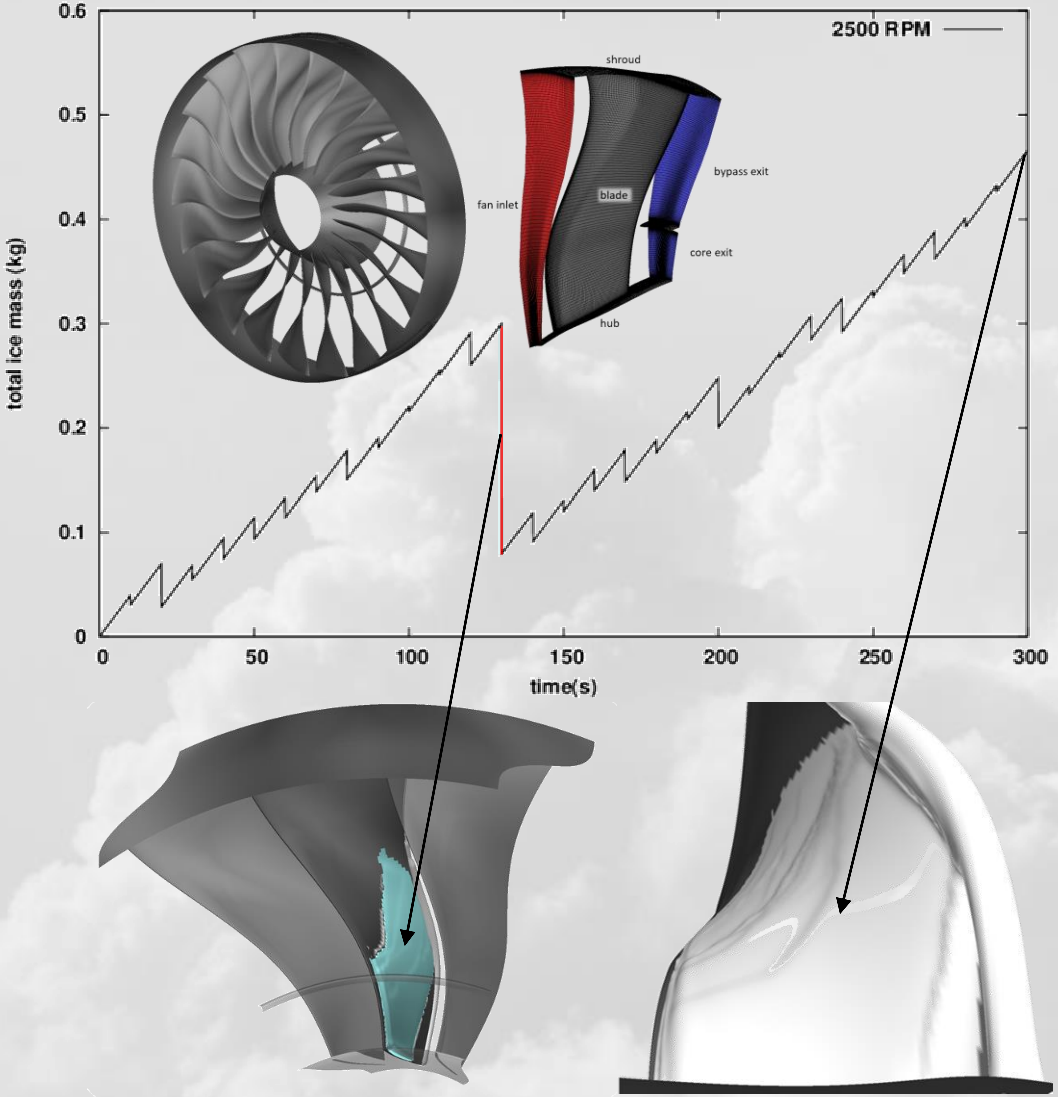


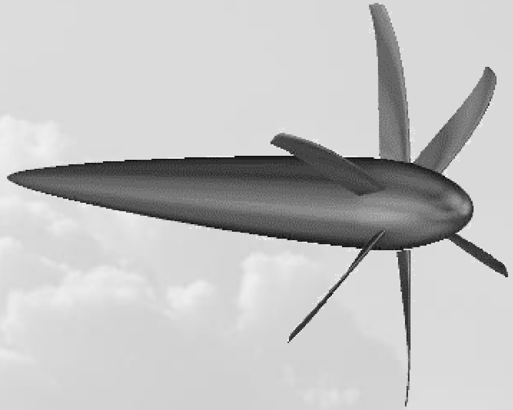
water film reinjection from propeller

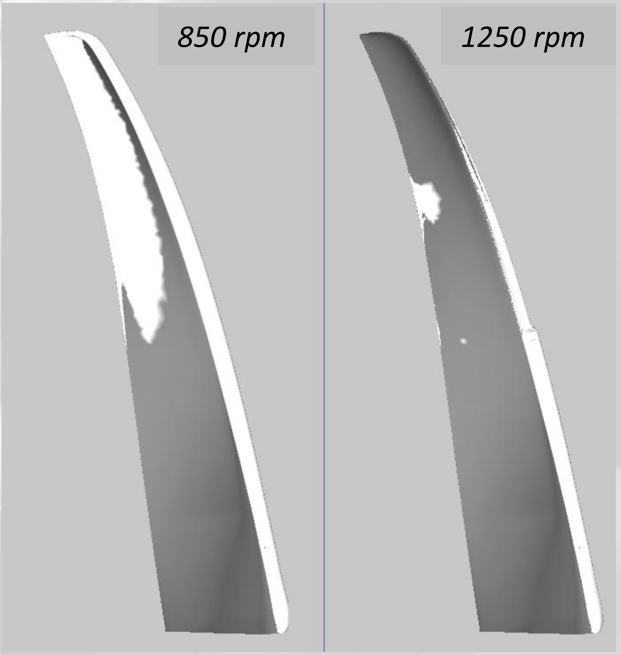
SAE 2019-01-2002

Ansys

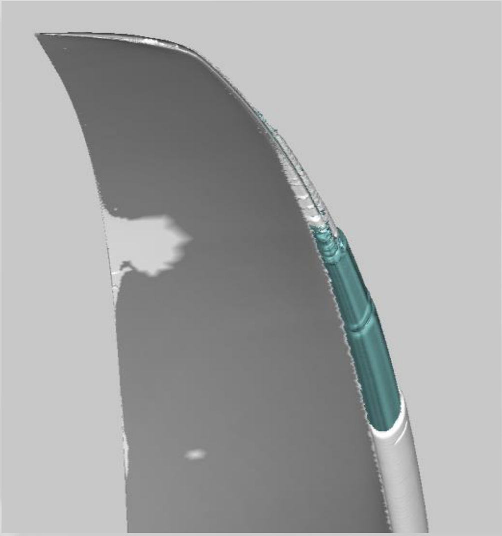
Ice shedding on rotating components





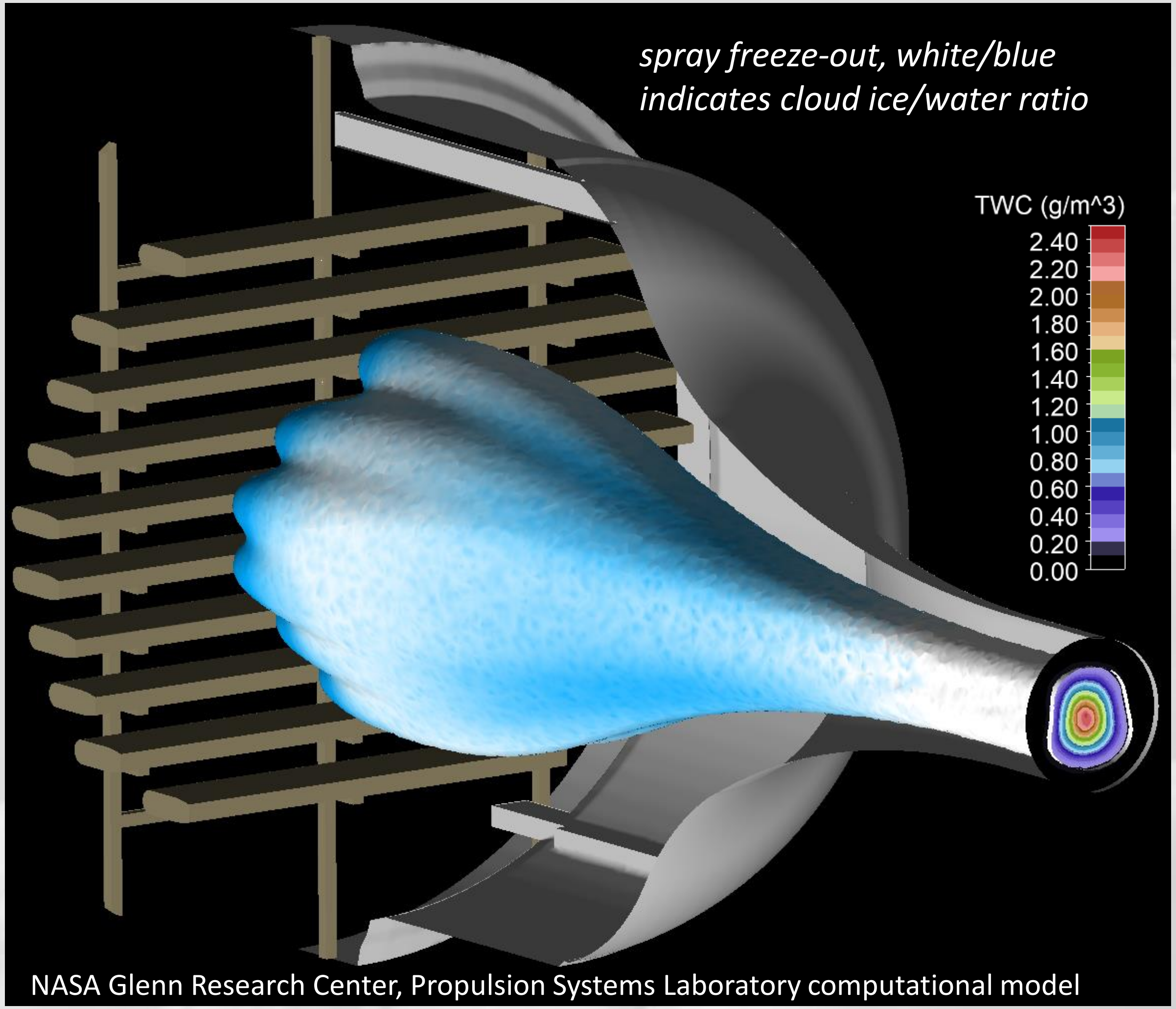


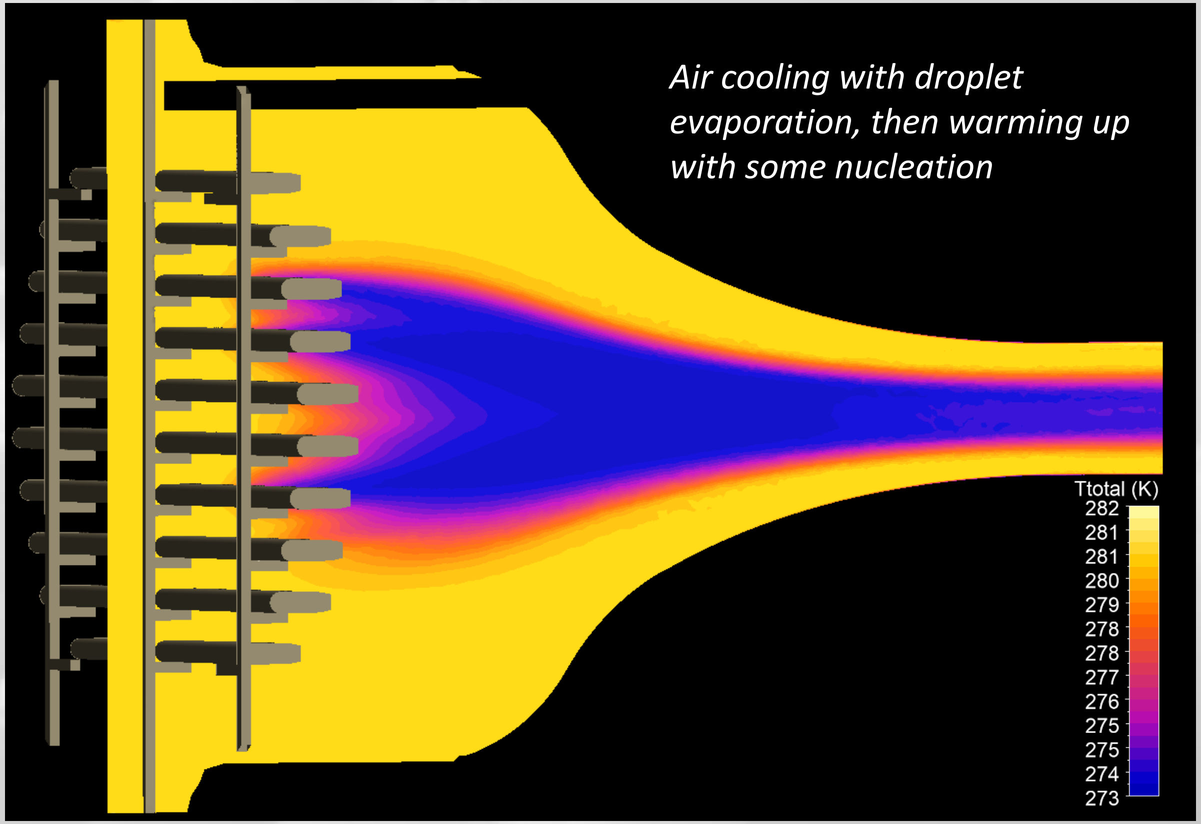
Ice shape at 30 mins



Largest shed piece

Droplet freeze-out: vapor, air, particle energy coupling





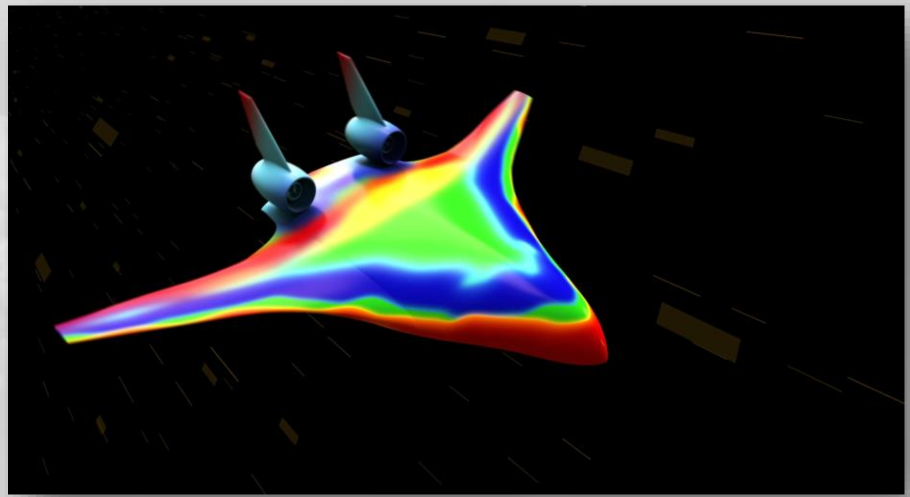
Concluding remarks

Concluding remarks

- +100000 flights / day, icing events rarely make the news
  - Decades of hard work, engineering and regulations
  - Systems put in place work well for the current generation of aircraft
- Why icing research?
  - **Sustainability**: next-gen aircraft will be more efficient, will require more efficient in-flight icing solutions
  - **Climate change**: Inclement weather patterns are increasing in intensity and frequency
  - **Electrification**: traditional bleed-air (hot-air)IPS that protect the majority of commercial jets will become irrelevant
  - **New materials:** increased use of composites instead of metal will limit the max T on protected zones
  - **Emerging IPS technologies**: Electro-mechanical, ultrasonic, hydrophobic/icephobic coatings
  - **Drones and UAVs**: much smaller aircraft, greater risk of icing, not enough real-estate to spare for IPS peripherals

Concluding remarks

- Why CFD?
  - Experiments need to be meticulously planned, not much room for do-overs
    - Wind tunnel experiments are a must, but waitlists are long (years if not months)
    - Flight testing requires a lot of prep, large ground crew, right weather conditions, in the right airspace...
  - CFD can help in
    - Planning experiments and flight tests
    - Reduce the test matrix
  - Engines are difficult to instrument for testing
    - Can be intrusive (cameras, probes)
    - CFD can reveal many details that are difficult to observe in experiments
  - AI/ML powered design optimization
    - Emerging technologies for product design
    - Require a lot of data to get smart
    - CFD can help populate this data very quickly



THANK YOU

Ansys